Evolving embodied intelligence

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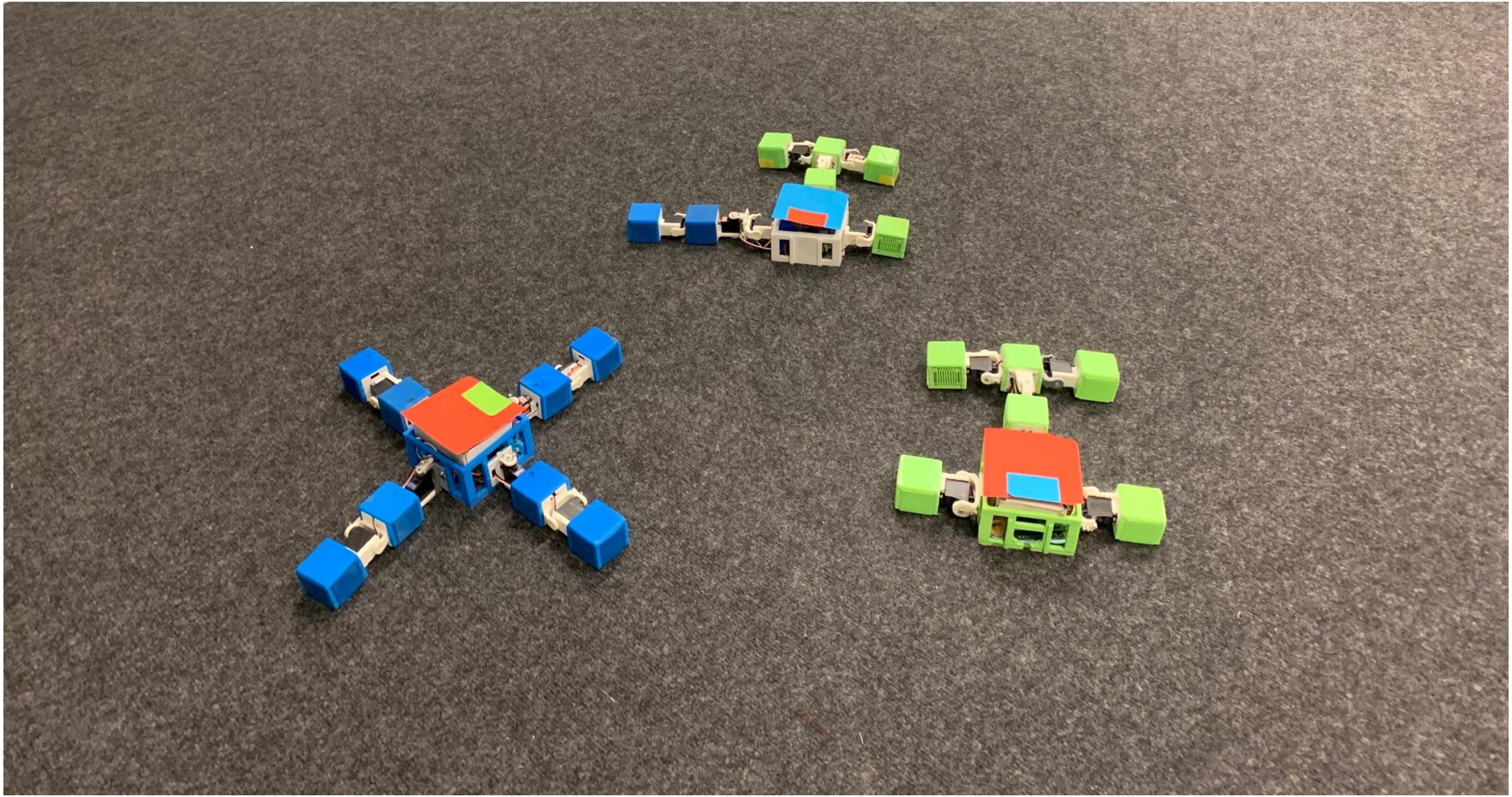
Vrije Universiteit Amsterdam, Netherlands
University of York, United Kingdom

Introduction to AI Course, VU, 07-10-2024

Primetime news on national television




NETWORK INSTITUTE
Triangle of Life

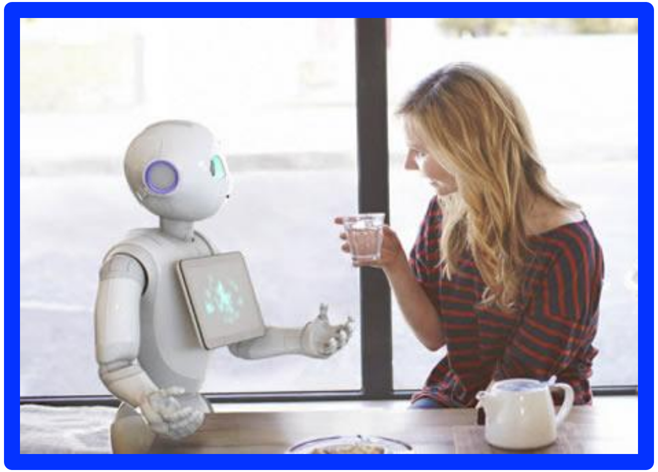
Evolution can create intelligence

Artificial evolution can create artificial intelligence

Can we evolve intelligence?

Digital AI

Embodied AI



machine translation

robot receptionist

chatbots

image analysis

warehouse robots

self-driving cars

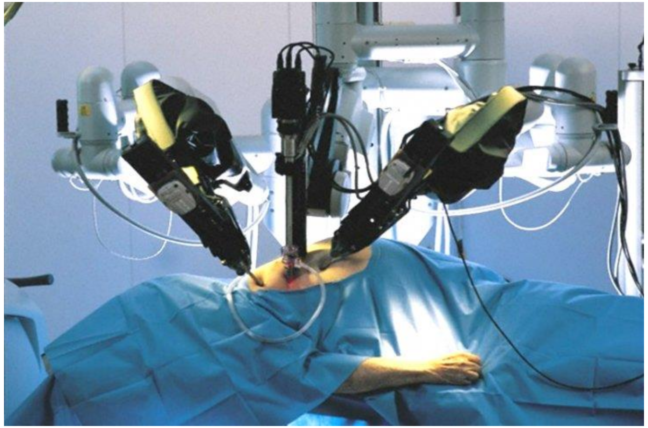
Thinking machines

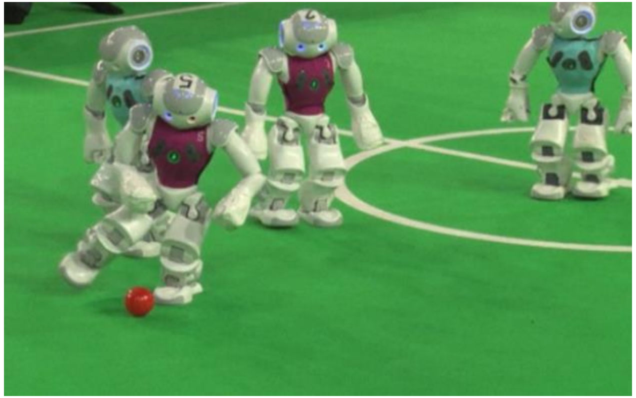
Acting machines


1997

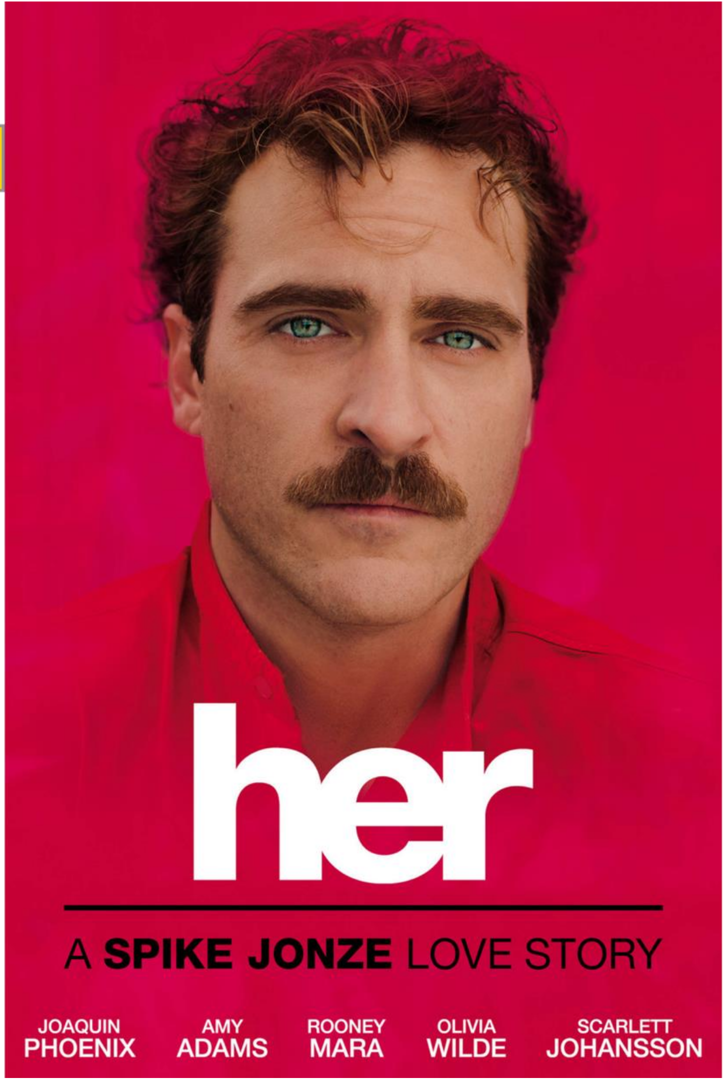

THINK
$300,000
$1,000,000
$200,000
KEN
WATSON
BRAD
2011

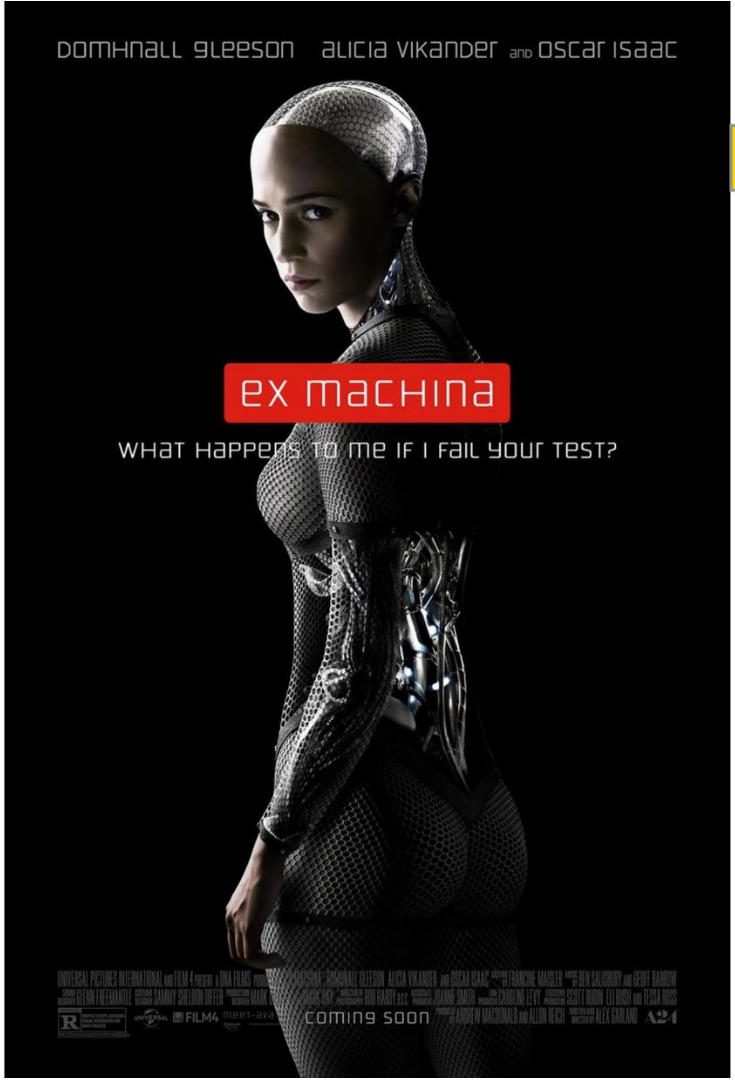

Google DeepMind
Challenge Match
8 - 15 March 2016
AlphaGo
2016
AlphaGo
Lee Sedol





8.0


her
A SPIKE JONZE LOVE STORY
JOAQUIN PHOENIX
AMY ADAMS
ROONEY MARA
OLIVIA WILDE
SCARLETT JOHANSSON

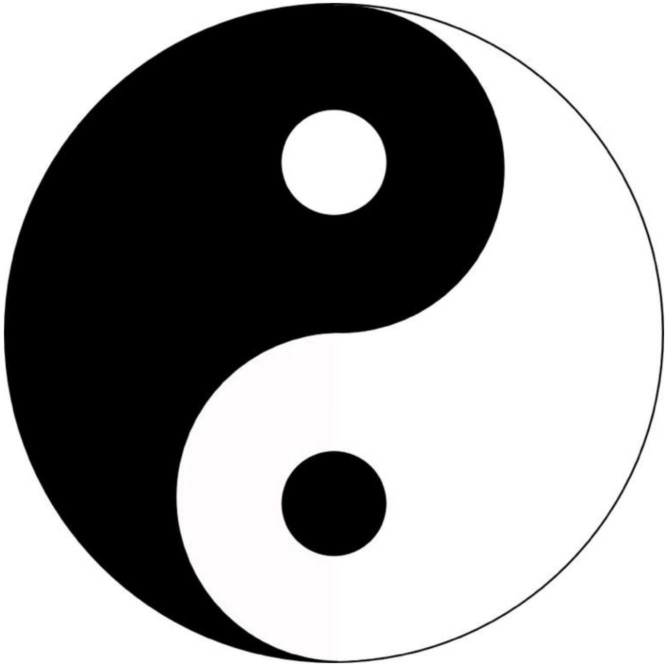

DOMHNALL GLEESON
ALICIA VIKANDER and OSCAR ISAAC
ex machina
WHAT HAPPENS TO ME IF I FAIL YOUR TEST?
coming soon

7.7

$$\text{Environment} + \underbrace{\text{body} + \overbrace{\text{brain}}^{\text{intelligence (narrow)}}}_{\text{intelligence (broad)}} \rightarrow \text{behavior}$$

Body

Brain



Hardware

Software

Optimal body = ?

Pragmatist

Antropo-chauvinist

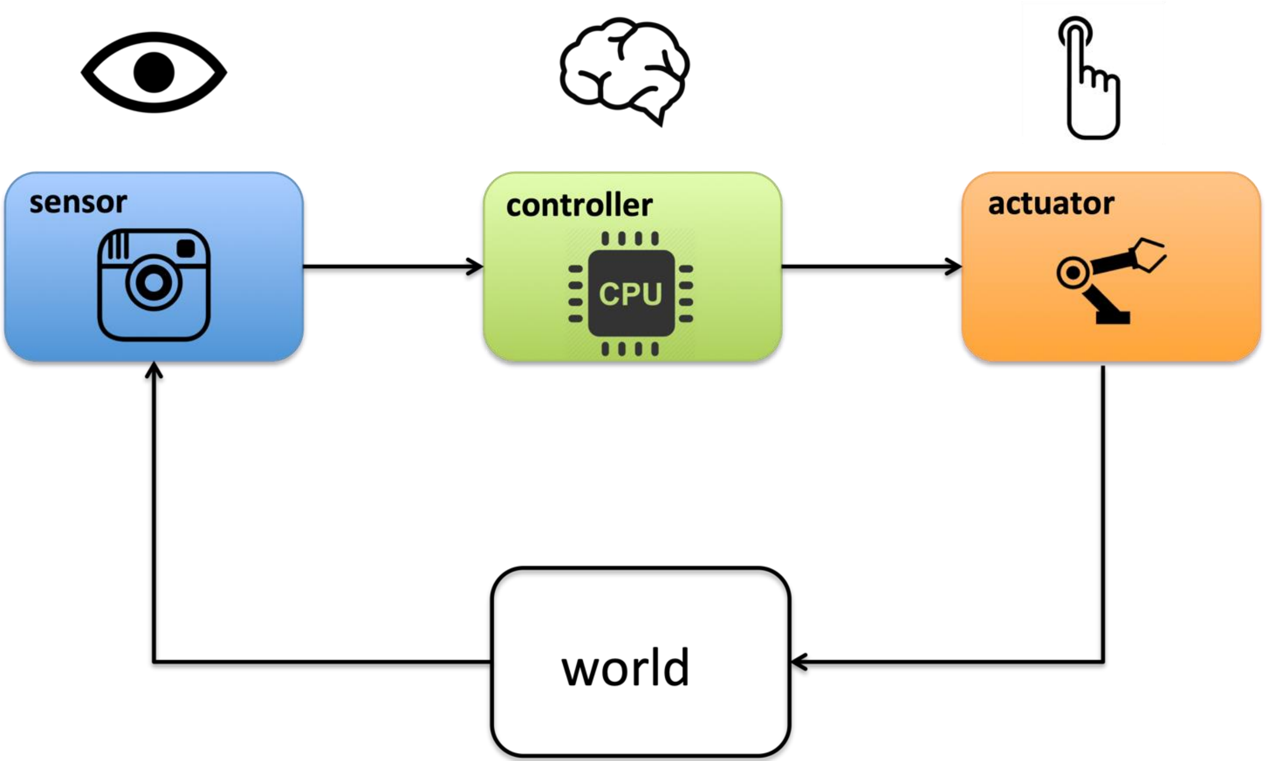
Radicalist

Optimal brain = ?

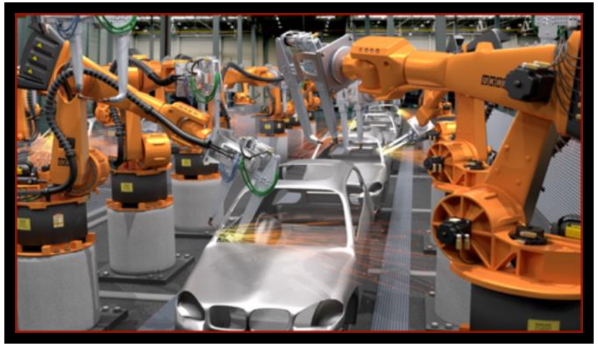
Can we evolve embodied intelligence?

Can we evolve robots?

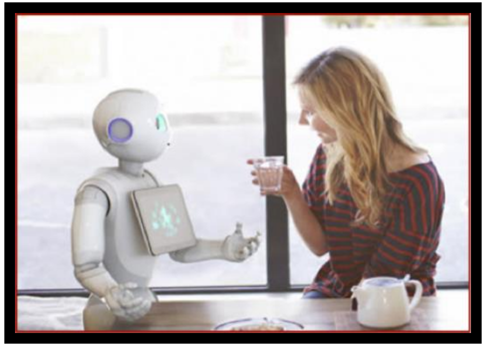
What is a robot?



stationary / industrial robot



humanoid robot

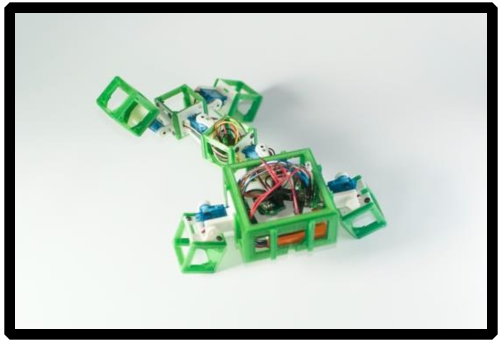


surgical (non-)robot

task-specific mobile robot

autonomous vehicle

academic research robot



What is a robot?

- Technically*: a machine that can 1) sense, 2) compute, 3) act
- Conceptually: a combination of
  - Hardware – morphology – body
  - Software – controller – brain
- Immersed in a sensory-motor loop / stream
- Operates in an environment – exhibits behavior
- Can be evaluated by task performance (if it is made for performing tasks)

* https://robots.ieee.org/learn/what-is-a-robot/

Artificial evolution?

Symbolic AI

good old-fashioned AI (GOFAI)

formal reasoning

top-down

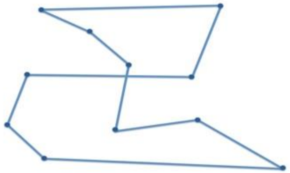
Sub-symbolic AI

modern, 21^{st} century AI

adaptation, learning, evolution

bottom-up

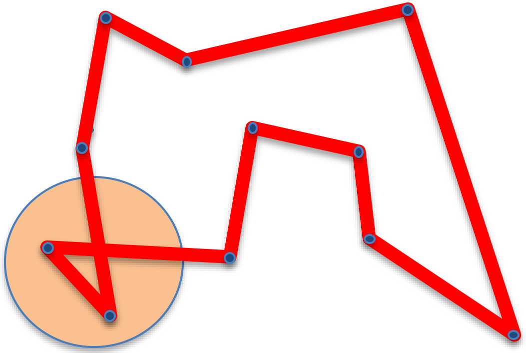
Individuals



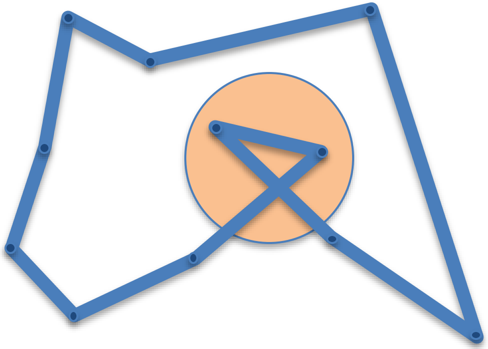
Selection

Reproduction

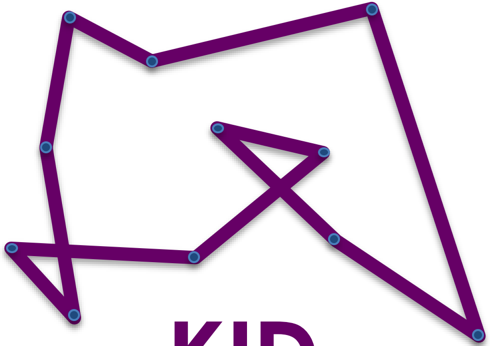
digital
sex



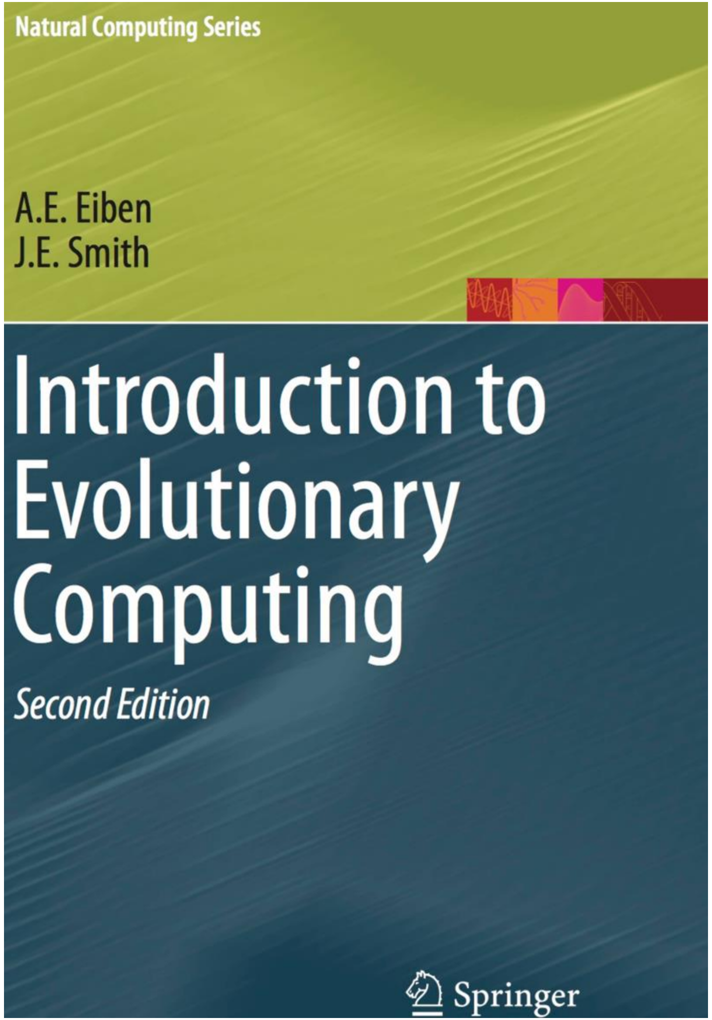
MOM



DAD



KID



Genetic algorithms

Evolution strategies

Evolutionary programming

Genetic programming

Differential evolution

…

Two pillars of evolution

Variation

Increasing population **diversity** by variation

- mutation
- recombination

Push towards **novelty**

Selection

Decreasing population **diversity** by selection

- of parents
- of survivors

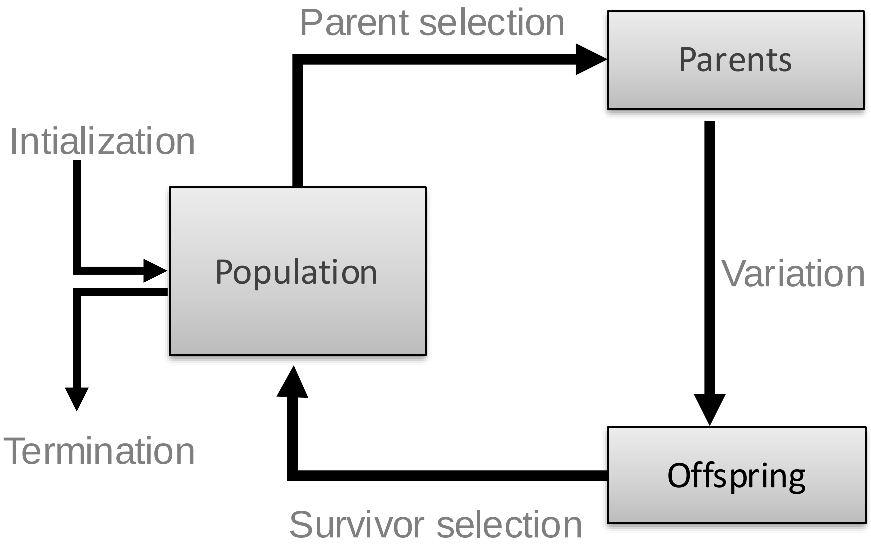
Push towards **quality**

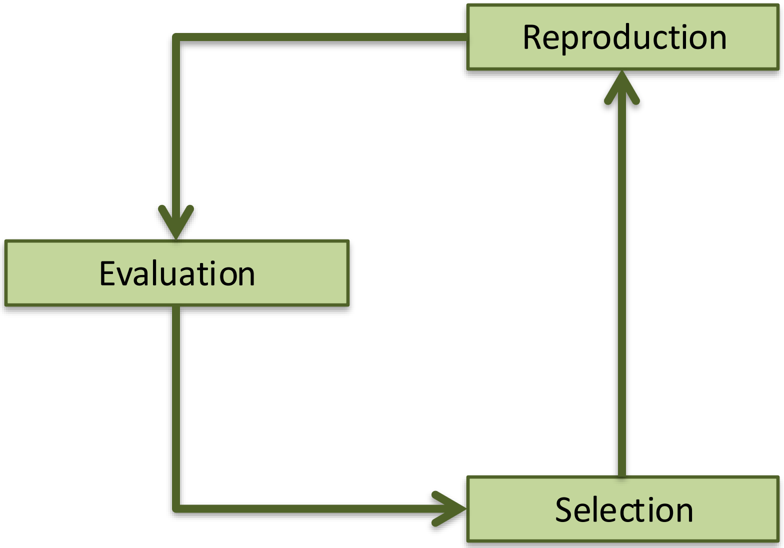
- Selection operators act on population level
- Variation operators act on individual level

Evolutionary Algorithms in pseudo-code

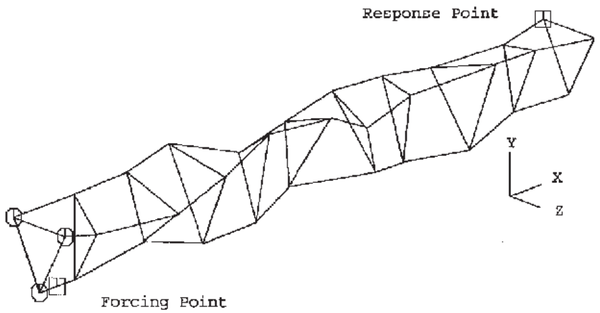
```
BEGIN
  INITIALISE population with random candidate solutions;
  EVALUATE each candidate;
  REPEAT UNTIL ( TERMINATION CONDITION is satisfied ) DO
    1 SELECT parents;
    2 RECOMBINE pairs of parents;
    3 MUTATE the resulting offspring;
    4 EVALUATE new candidates;
    5 SELECT individuals for the next generation;
  OD
END
```

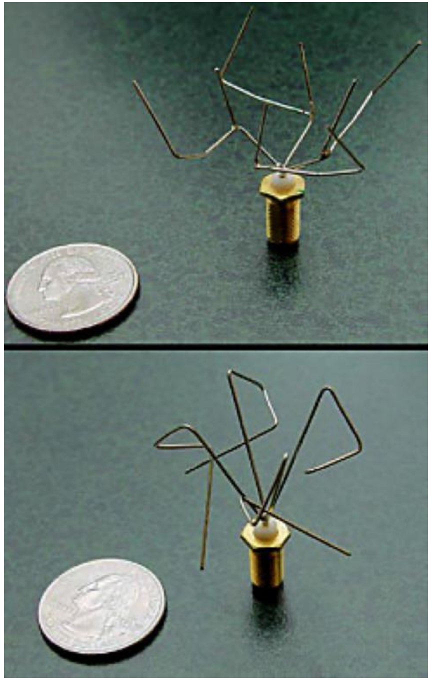
Evolutionary Algorithms diagram

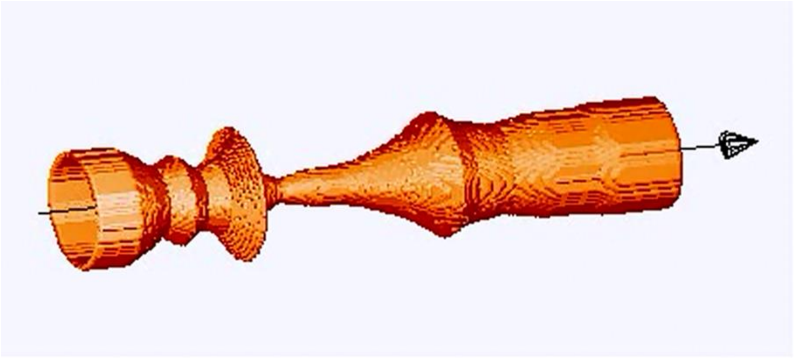




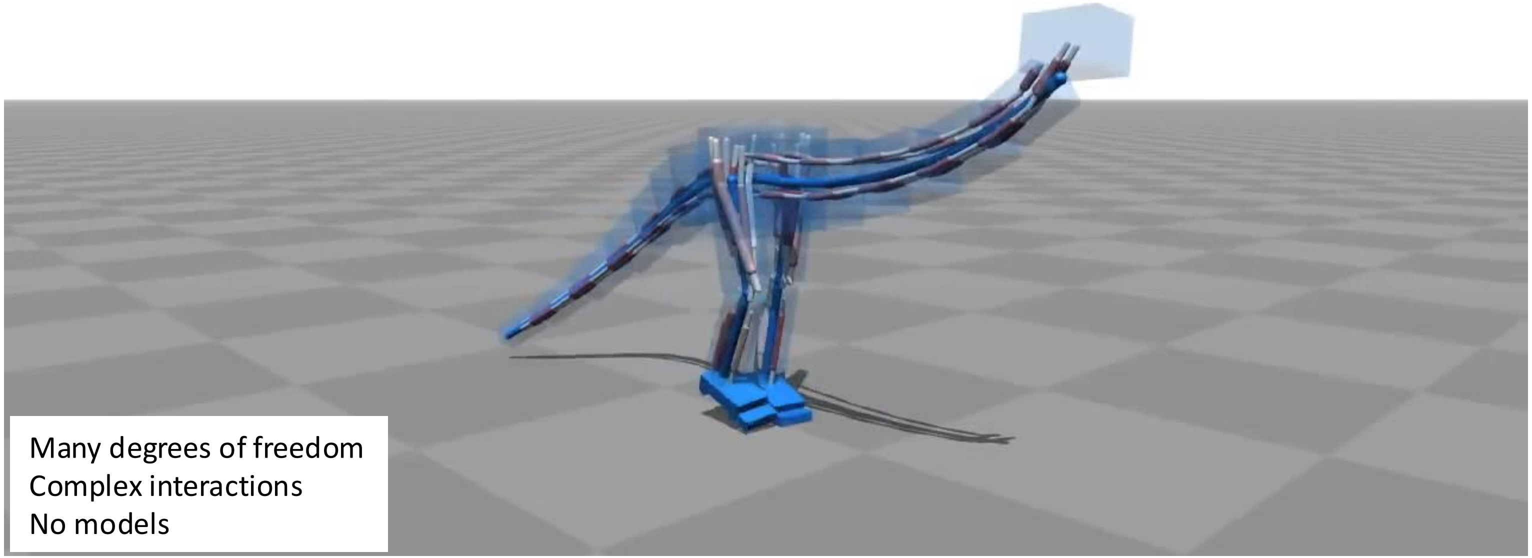
Evolutionary design



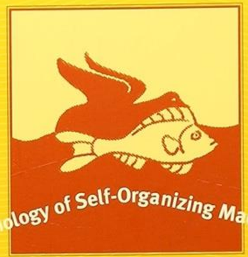




Geijtenbeek et al., Flexible Muscle-Based Locomotion for Bipedal Creatures, ACM ToG, 2013



Evolutionary Robotics



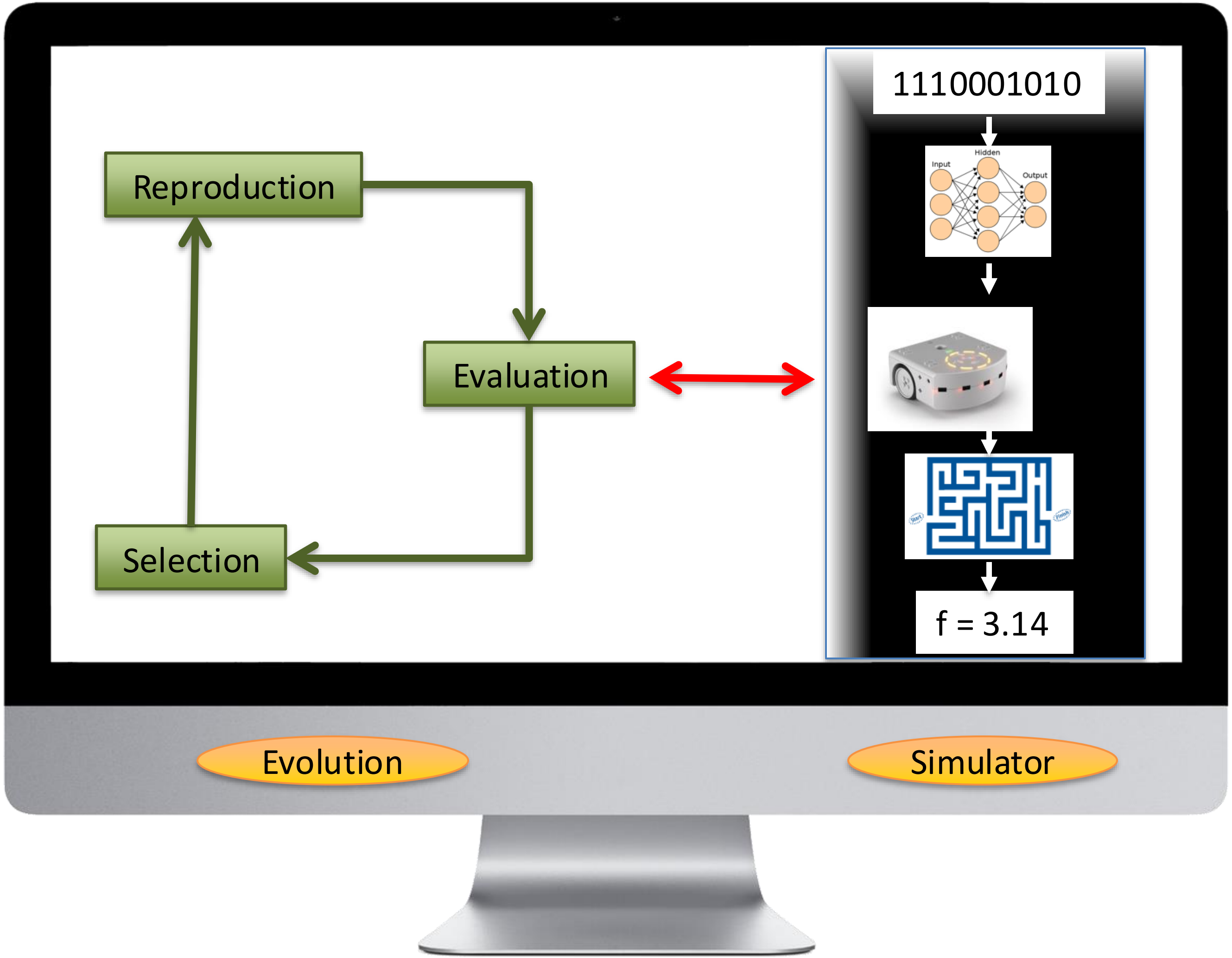
The Biology, Intelligence, and Technology of Self-Organizing Machines

Stefano Nolfi *and* Dario Floreano

Evolutionary Robotics

"Aims to apply evolutionary computation techniques to evolve the overall design or controllers, or both, for real and simulated autonomous robots" (Vargas et al., 2014)"

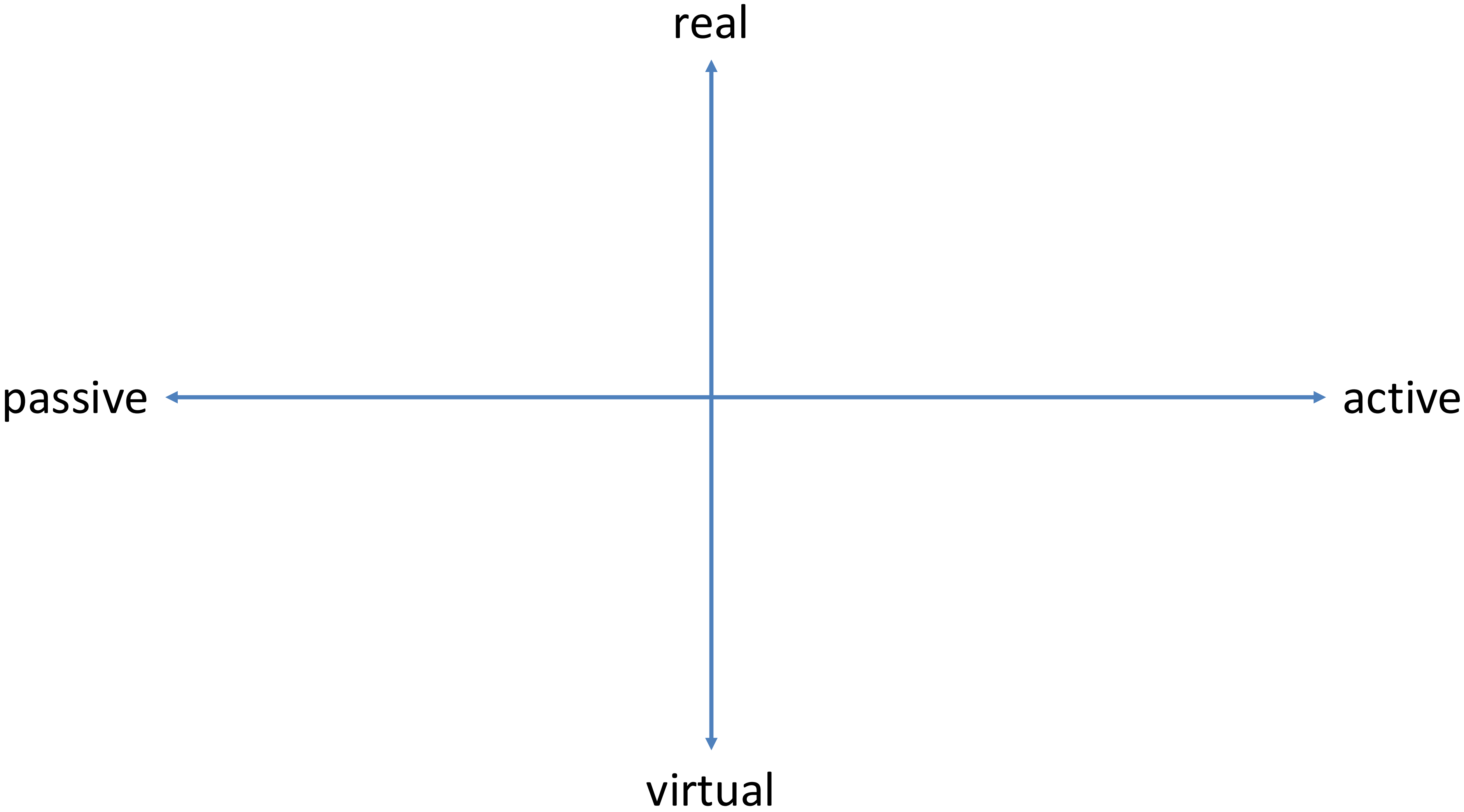
2nd order engineering

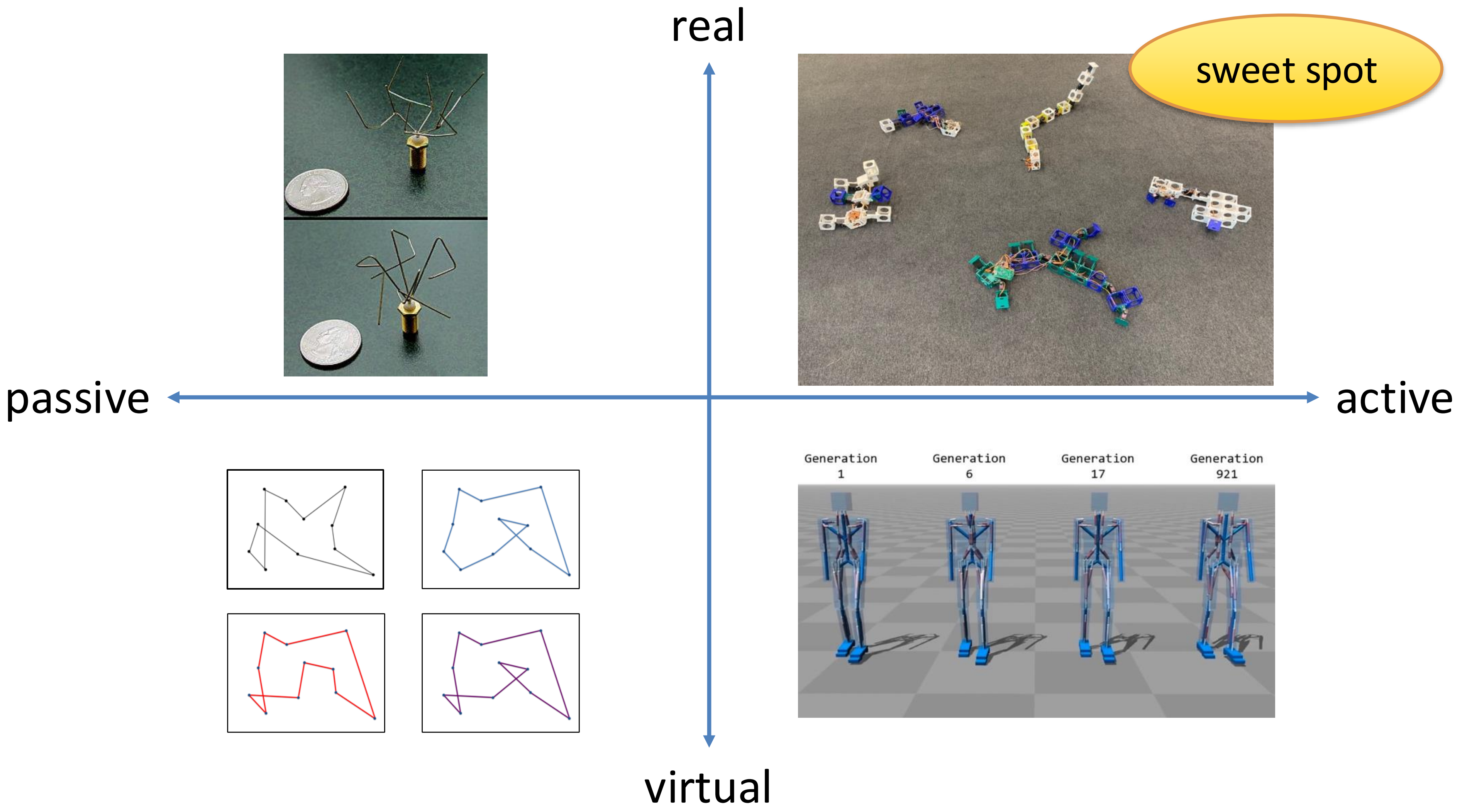


Reproduction
Evaluation
Selection
1110001010
Hidden
Input
Output
f = 3.14
Evolution
Simulator

Matter matters

REALITY GAP

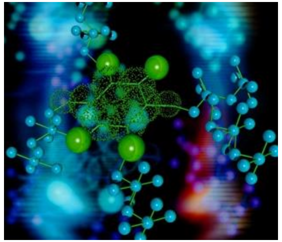

real
passive
active
virtual


real
sweet spot
passive
active
Generation 1
Generation 6
Generation 17
Generation 921
virtual

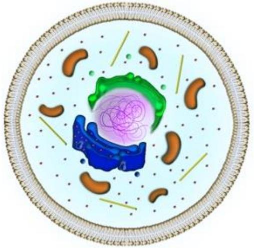
A.E. Eiben, Tech Kangaroos - Evolution at Work, TEDx Danubia, 2011
https://www.youtube.com/watch?v=WJX_wAKhg8A

Embodied Artificial Evolution

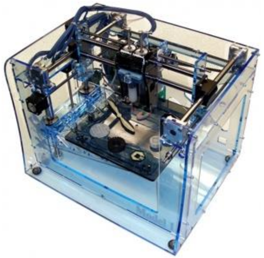
Eiben, Kernbach, Haasdijk, Embodied artificial evolution: Artificial evolutionary systems in the 21st Century, Evolutionary Intelligence, 5(4):261-272, 2012



Chemistry



Biology



Engineering

Eiben, Kernbach, Haasdijk, Embodied artificial evolution: Artificial evolutionary systems in the 21st Century, Evolutionary Intelligence, 5(4):261-272, 2012

GRAND CHALLENGES

1. Body types
2. How to start (reproduction)
3. How to stop (kill switch)
4. Evolvability & speed
5. Process control & methodology
6. Body+brain evolution + learning

Eiben and Smith, From evolutionary computation to the evolution of things, ***Nature***, 2015

20th century

21st century

WETWARE
in vivo

SOFTWARE
in silico

HARDWARE
in materio

Biosphere

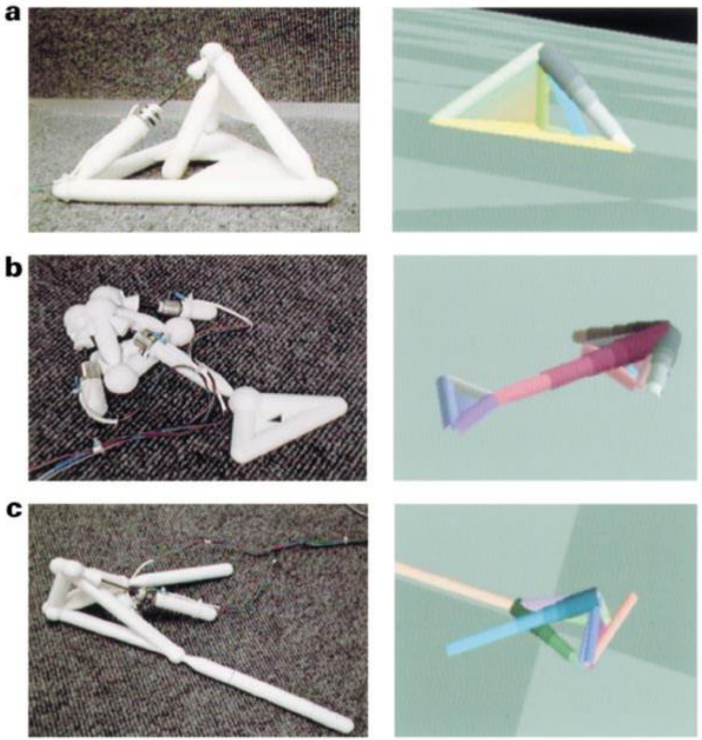
Evolutionary Computing

Evolution of Things

Lipson, Pollack, Automatic design and manufacture of robotic lifeforms, *Nature* 406, 2000

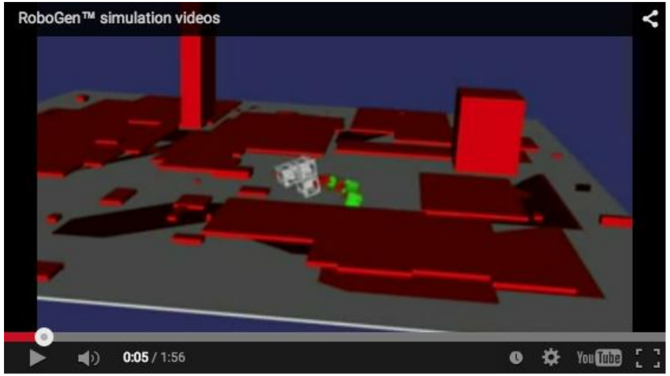
- GOLEM project
- EA + simulator
- Evolution of body + controller
- Evolved robot fabricated afterwards

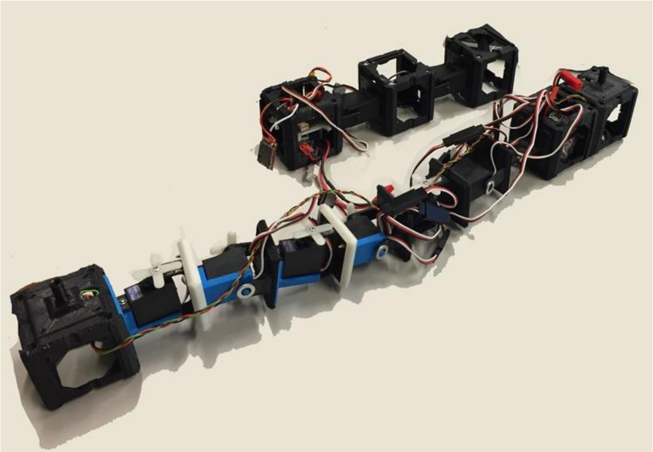
- Off-line evolution
- Evolution in simulation
- No sensors – are they robots?



Auerbach *et al.*, RoboGen: Robot generation through artificial evolution, Proc. of Artificial Life 2014, MIT Press, 2014

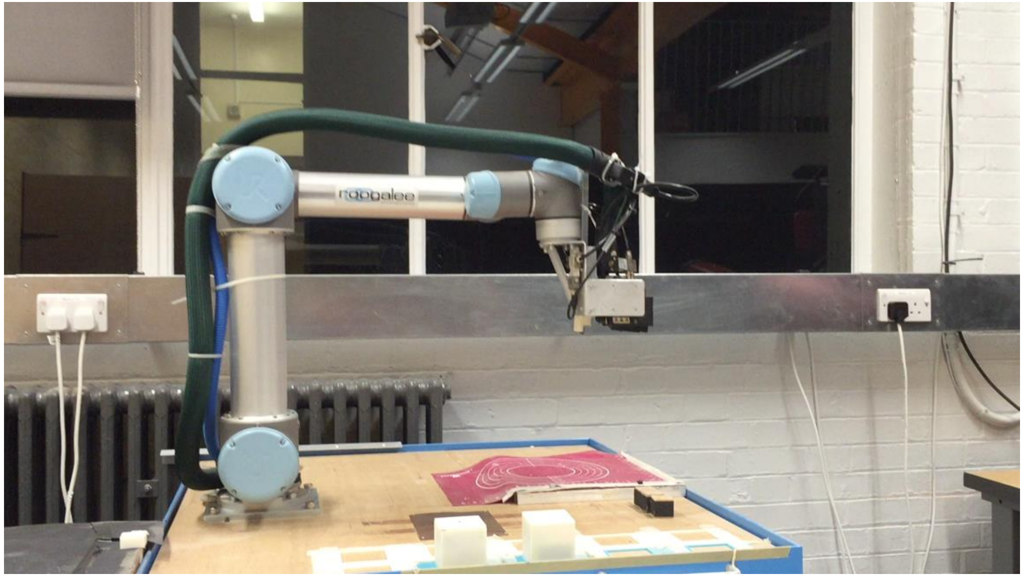
- Didactic tool and experimental testbed
- Open-source SW & HW platform
- Robots:
  - Fixed parts, 3D-printed parts
  - Arduino board, NN controllers
  - Tree-based genetic code
- Evolution
  - Off-line, centralized, supervised, simulated
  - Fitness evaluation in isolation
- End product of simulated evolution is constructed in HW <u>by hand</u>





Brodbeck, Hauser, and Iida, Morphological evolution of physical robots through model-free phenotype development, PLoS One, 10(6) 2015.

- Evolution
  - Off-line, centralized, physical trials
  - Fitness evaluation in isolation
  - Only mutation, no crossover
- Each individual is constructed in HW
  - By “mother robot”
  - Hands-free in Experiment 1
  - Human-assisted in Experiment 2
- No sensors – are they robots?



Essential question?

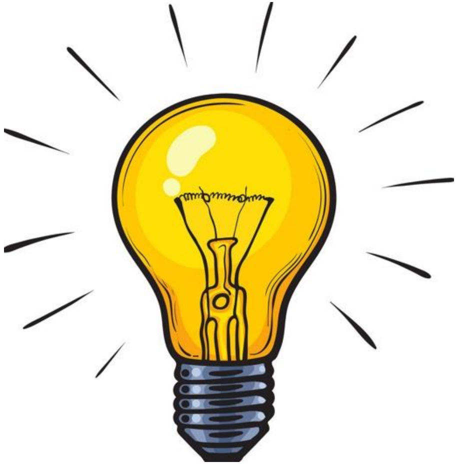
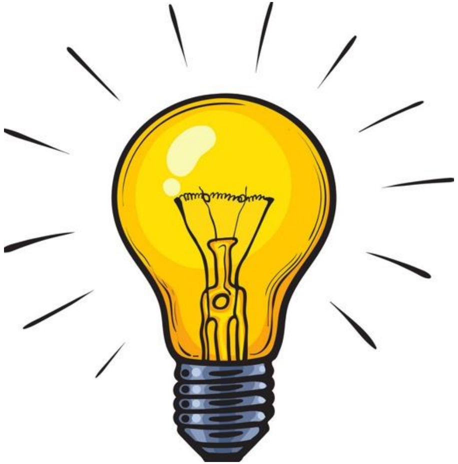

how i
met your
mother

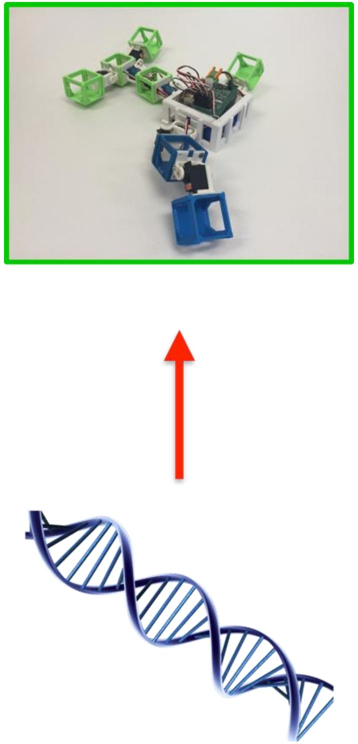
This is a good moment for a break

Can we evolve real robots?

selection 

reproduction 




LULZBOT
TAZ

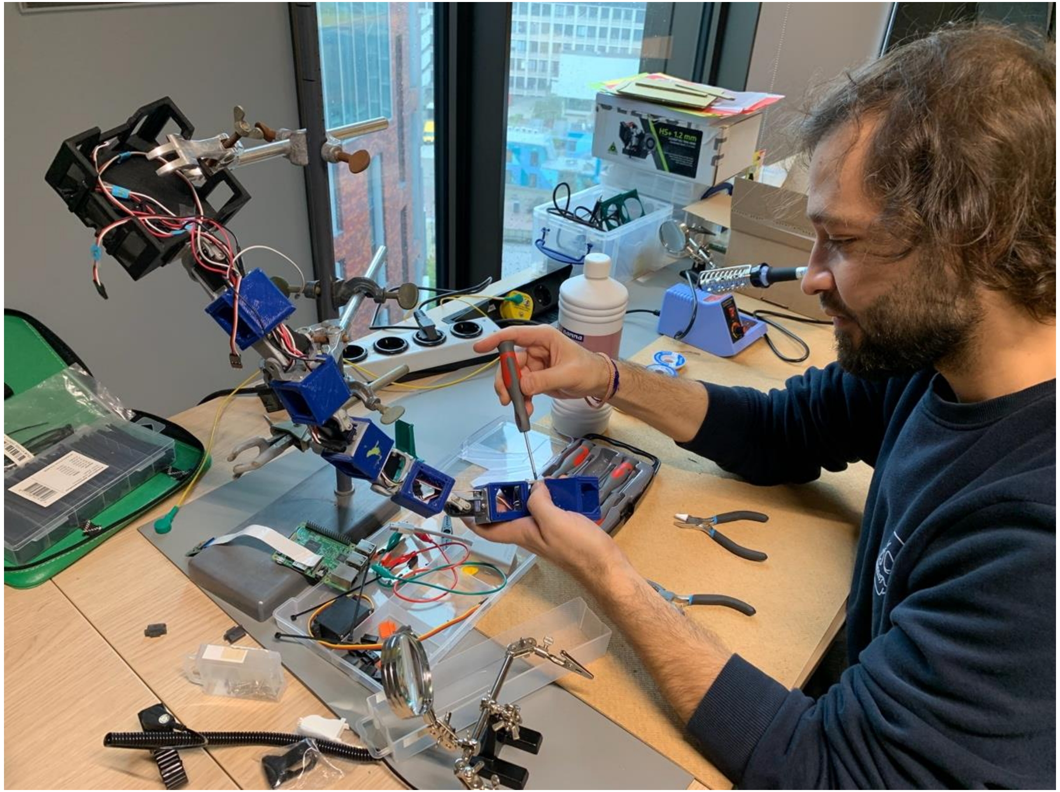
Genetic code

Head_1 = 32, 81, 5
Wheel_1 = A, 1
Wheel_2 = A, 41
Wheel_3 = B, C, 3.141
Camera = rot(29)
WiFi = comm_98
Body = M[block_1, 2 ;
block_2, 4,5 ;
block_3, 21 ;
block_6, 8 ;
cilinder(4)]
Light = R / G / B, LED*
Mic = sound_x
Leg_1 = [front, 0.4, 1.2, steel]
Leg_2 = [rear, 0.9, 2.5, TPU]
. . . .

Genotype

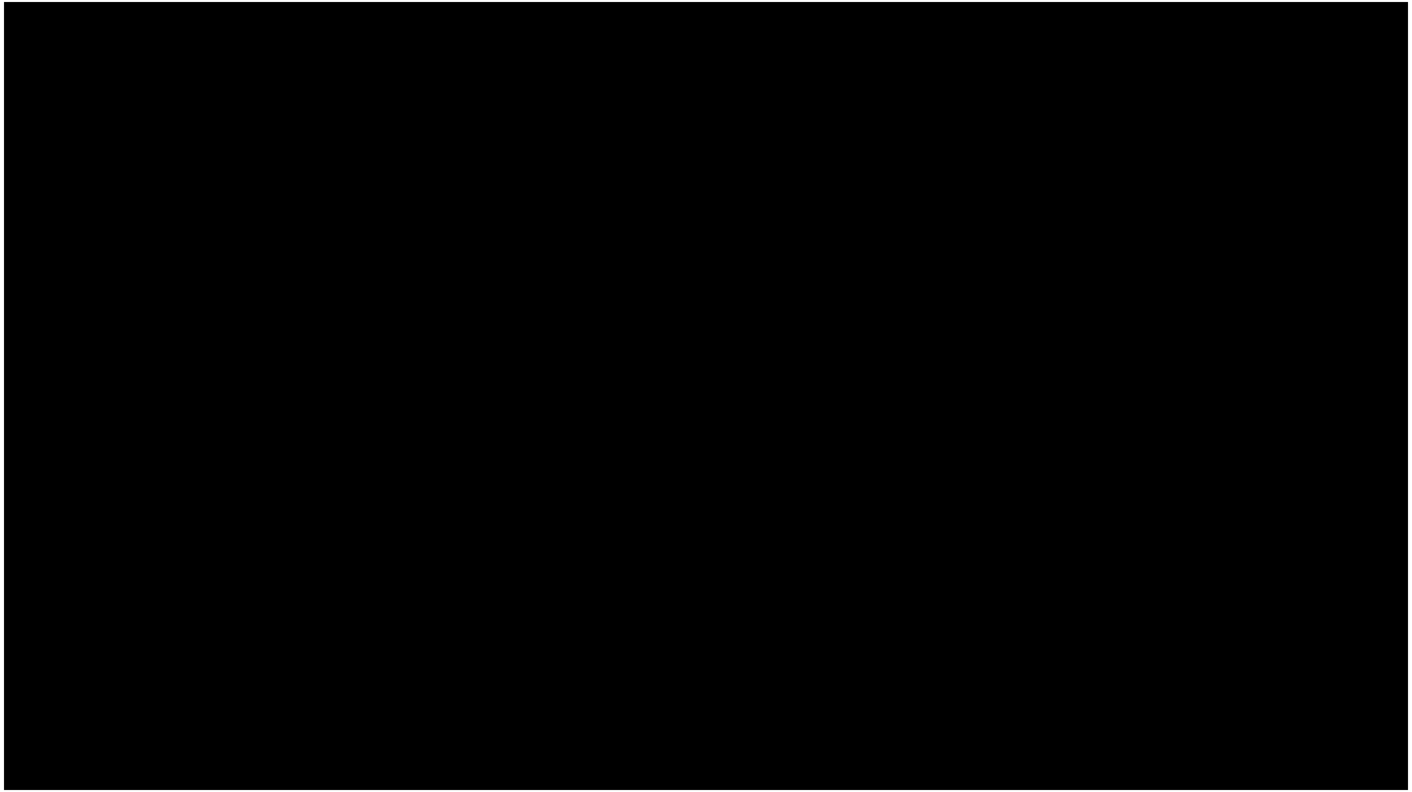
Phenotype

Birth

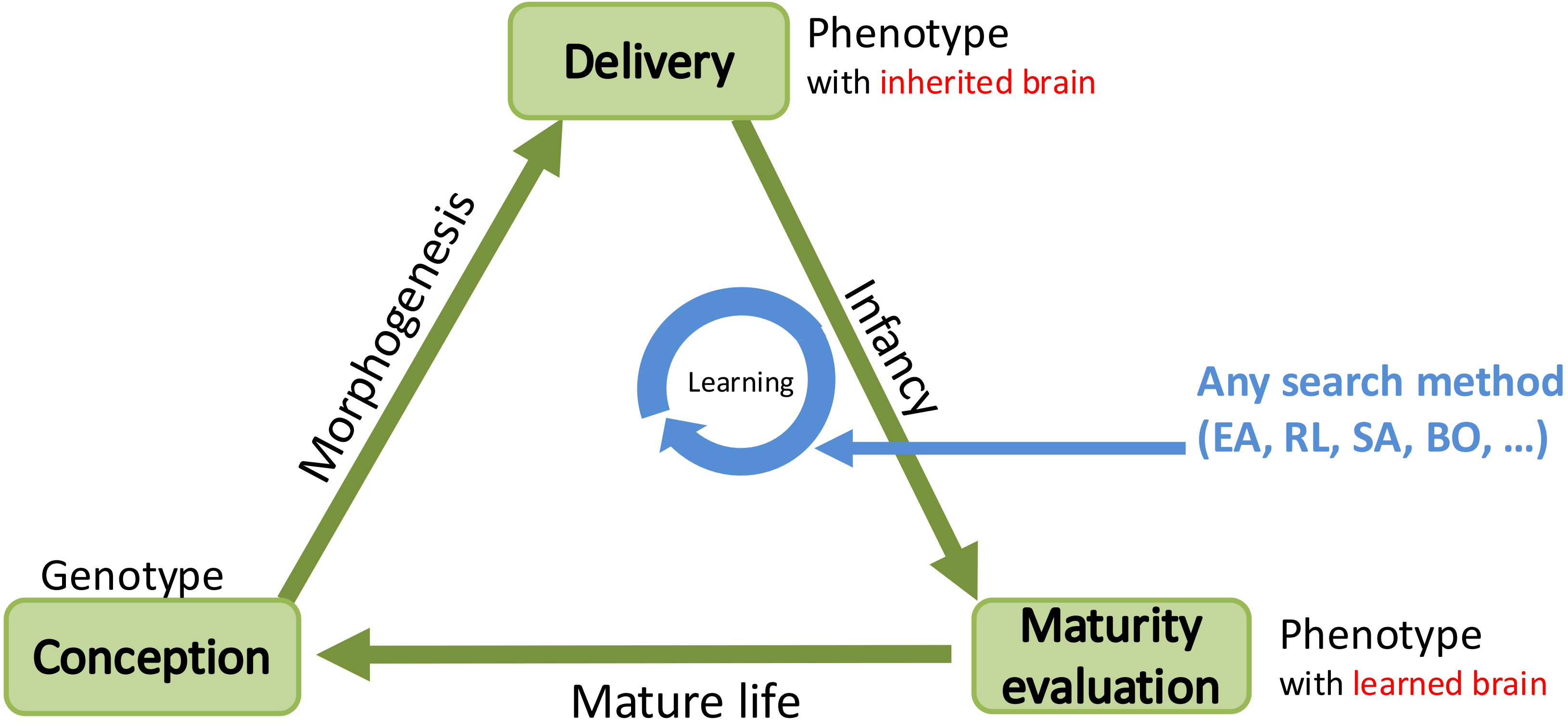


Jelisavcic, De Carlo, Hupkes, Eustratiadis, Orlowski, Haasdijk, Auerbach, Eiben, Real-World Evolution of Robot Morphologies: A Proof of Concept, ***Artificial Life 23(2),*** 2017.

Full movie (5min): https://www.youtube.com/watch?v=BfcVSb-Q8ns



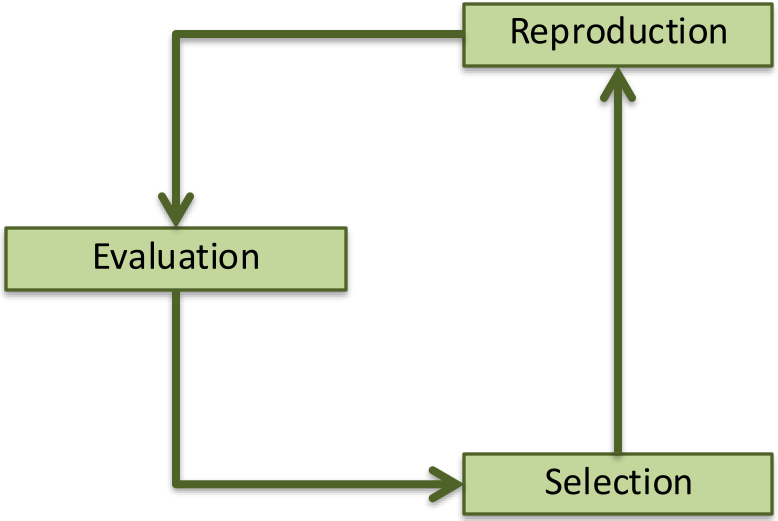
Eiben, Bredeche, Hoogendoorn, Stradner, Timmis, Tyrrell, Winfield, The Triangle of Life: Evolving Robots in Real-time and Real-space, ECAL, 2013

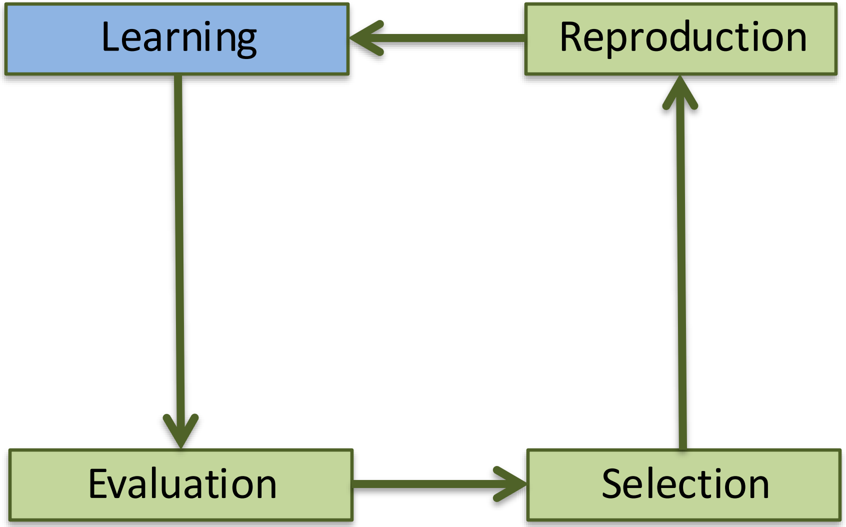


Not only for real-world robot evolution

For simultaneous body + brain evolution
simulated or real

Eiben and Hart, If it evolves it needs to learn, ***GECCO workshop***, 2020





Off-line evolution

before the job

On-line evolution

on the job

Design stage → Deployment → **Operational stage**

Breeding scenario

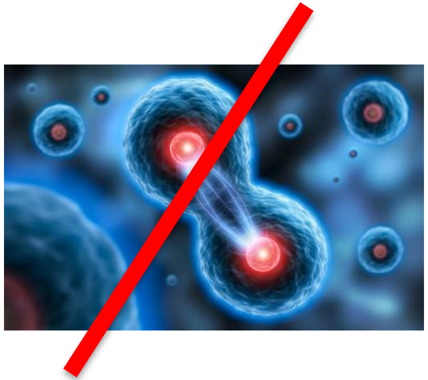
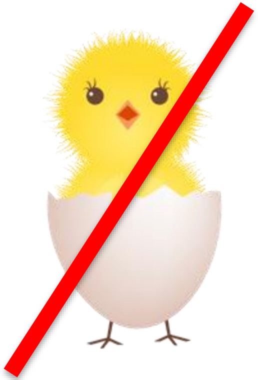
optimization

Mars scenario

adaptation

~~Can~~

Should we evolve real robots?


Fundamental insight:

not necessary

Ethical constraint:

not desirable

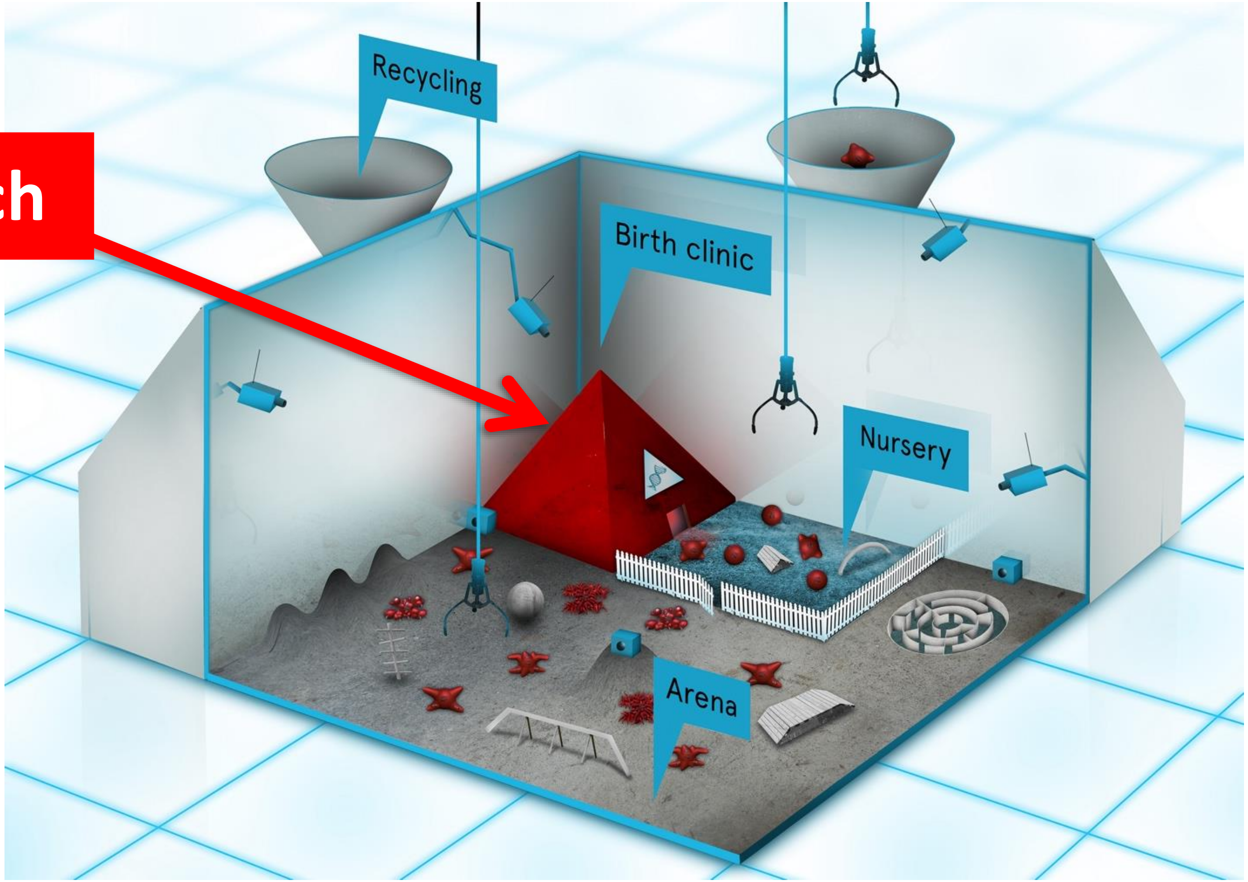
~~**Distributed (self-)reproduction**~~

Centralized, externalized reproduction

Does not invalidate the concept of evolution

Eiben, EvoSphere: the world of robot evolution, *TPNC conference*, 2015.

Kill switch

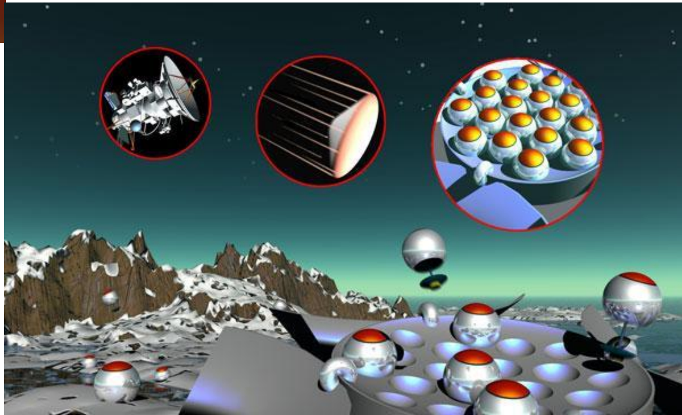


Applications

Rain forest monitoring

Medical nano robots

Terraforming



laying the groundwork

developing know-how

seeking deeper understanding

(patterns, relationships, invariants)

Evolution of intelligence

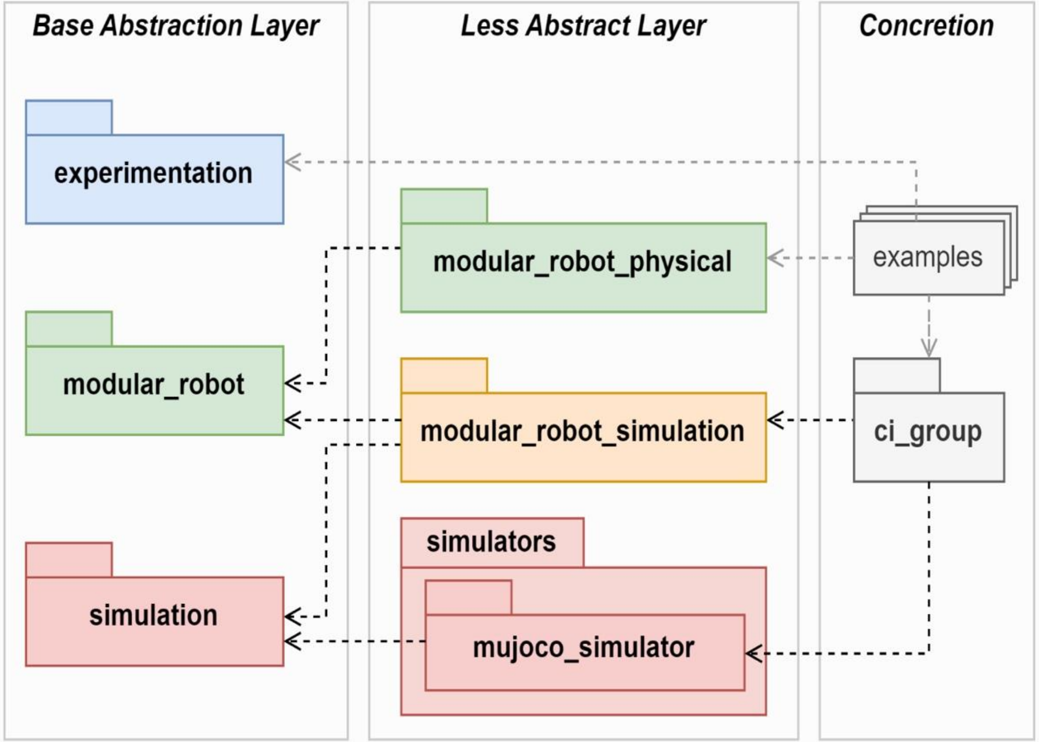
preconditions, attractors, invariants

Our simulator: Revolve (**R**obot **EVOLVE**)

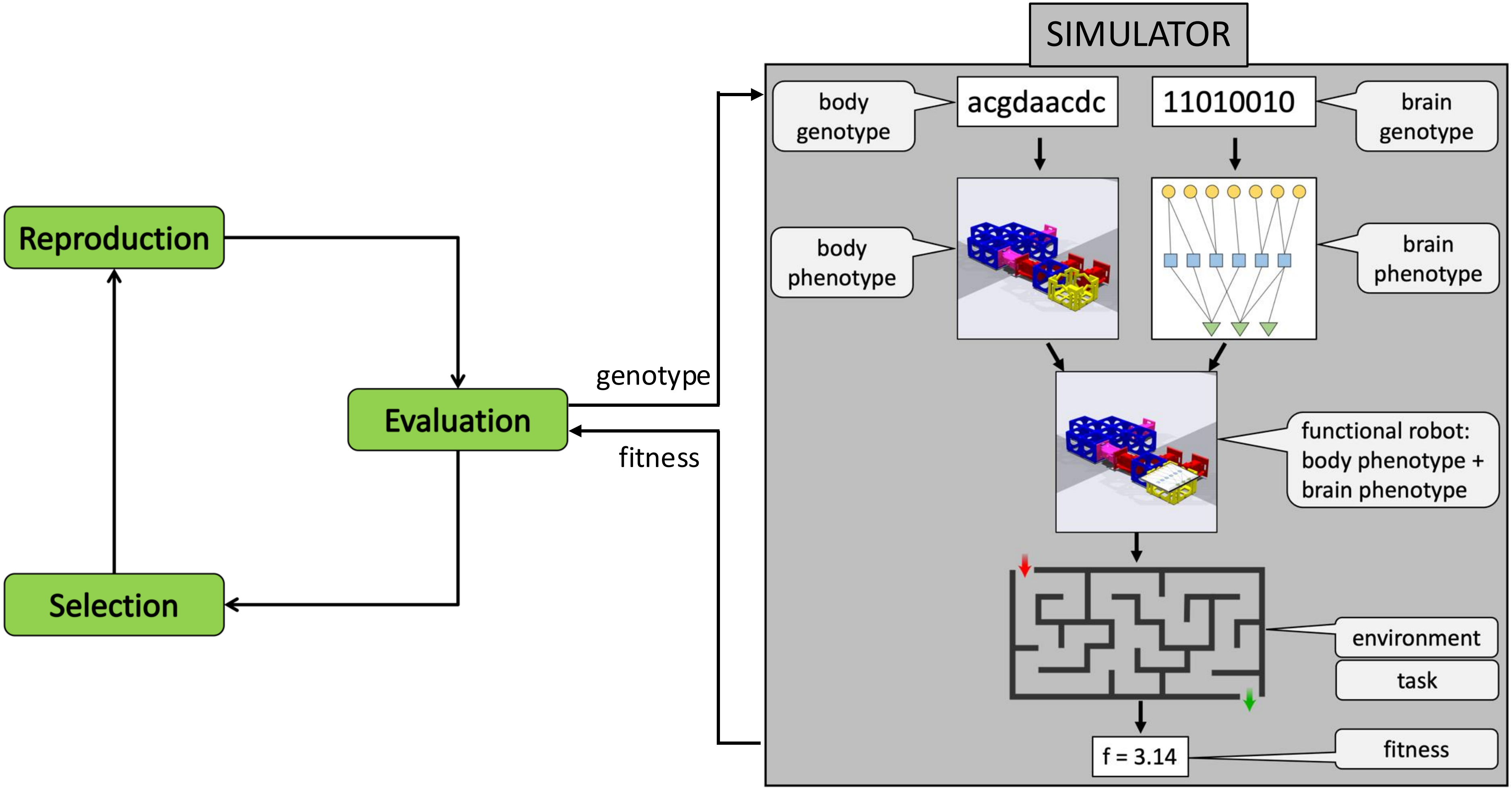
- https://ci-group.github.io/revolve2/
- MuJoCo based, Python
- Modular robots after the RoboGen system
- Several options for evolutionary and learning methods
- Several options for environments and tasks
- Several tools and scripts for analysis
- Simulated & real world, digital and physical twins
  - "everything we can simulate, we can build"
  - "everything we can build, we can simulate"

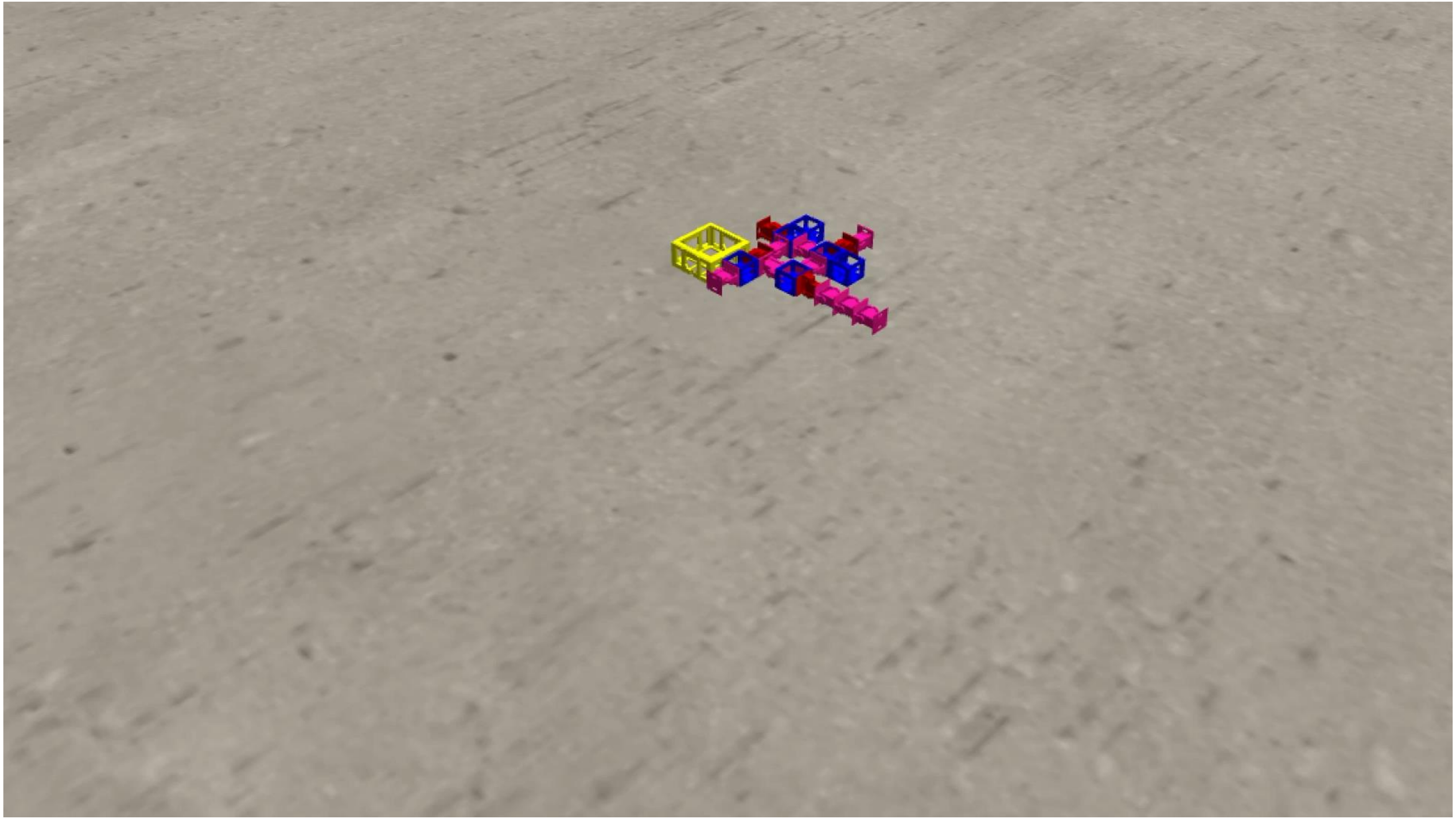
Revolve2 documentation

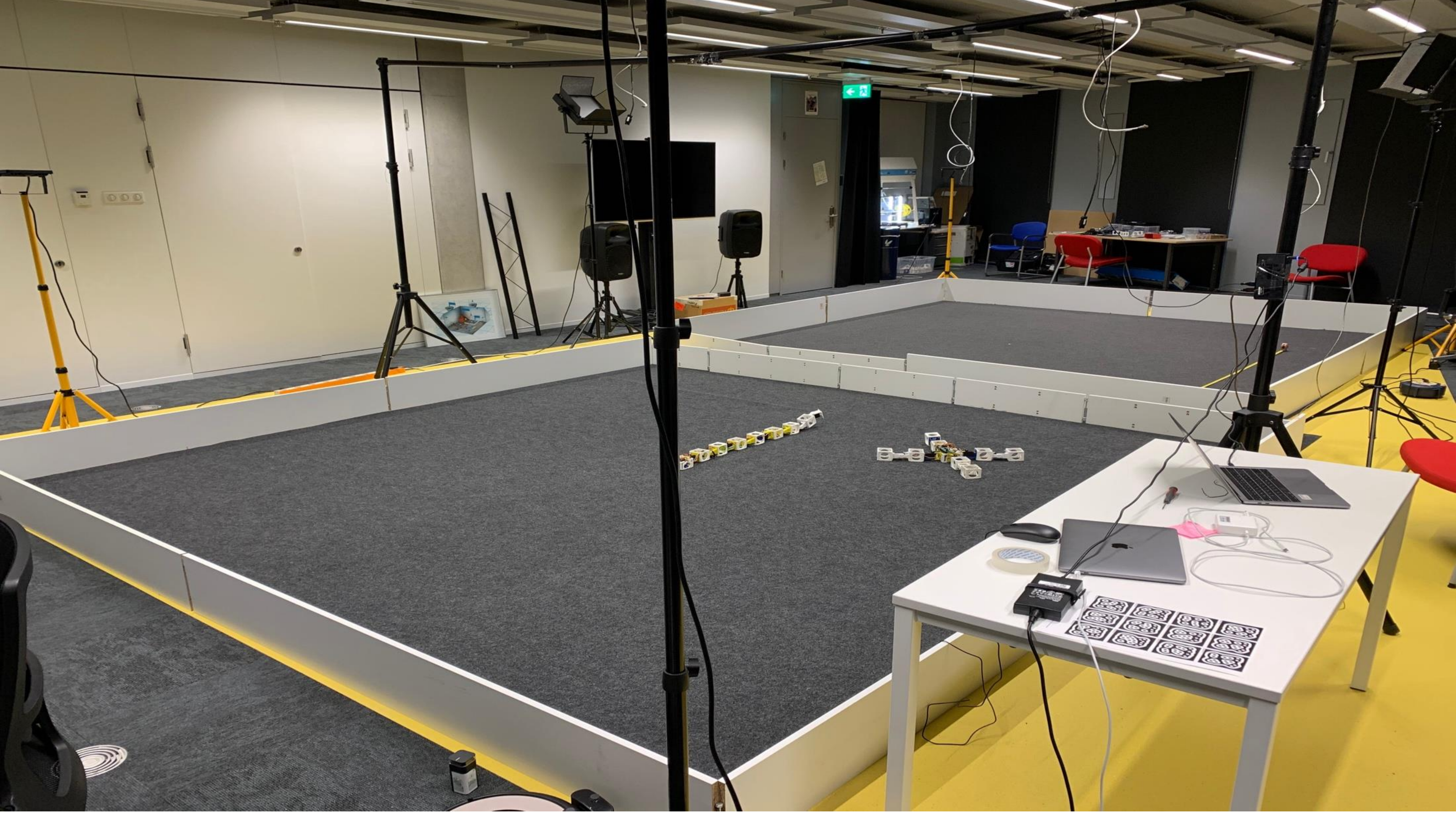
Revolve2 is a collection of Python packages used for researching evolutionary algorithms and modular robotics. Its primary features are a modular robot framework, an abstraction layer around physics simulators, and evolutionary algorithms. The structure of the packages in Revolve2 is as follows:

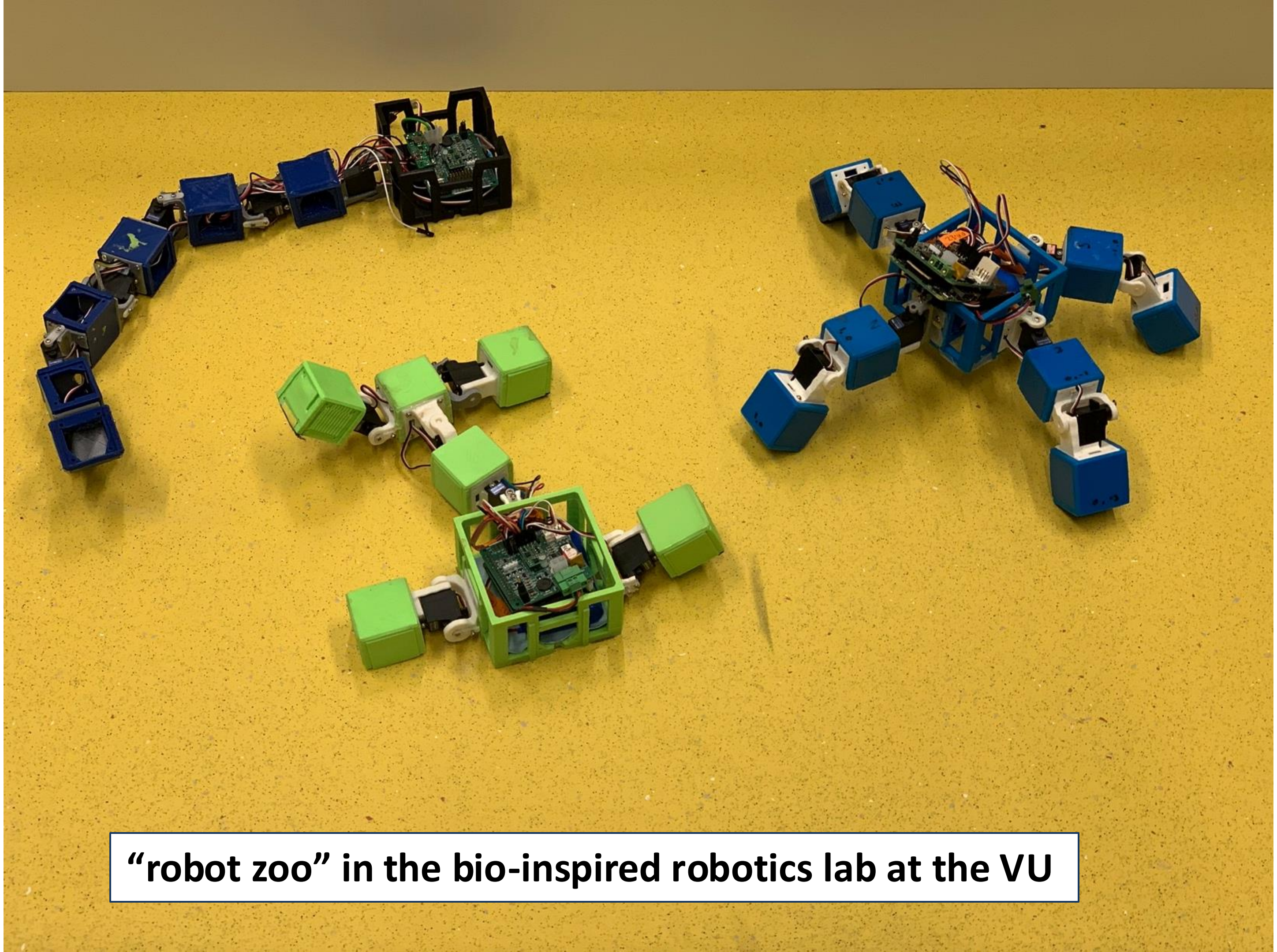


Revolve2's code resides at https://github.com/ci-group/revolve2.

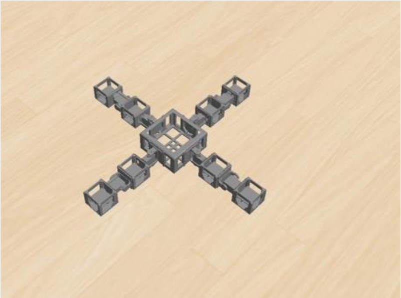

Reproduction
Evaluation
Selection
genotype
fitness
SIMULATOR
body genotype
acgdaacdc
11010010
brain genotype
body phenotype
brain phenotype
functional robot: body phenotype + brain phenotype
environment
task
f = 3.14
fitness

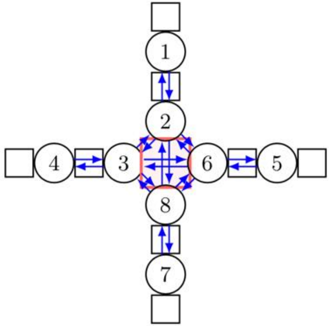


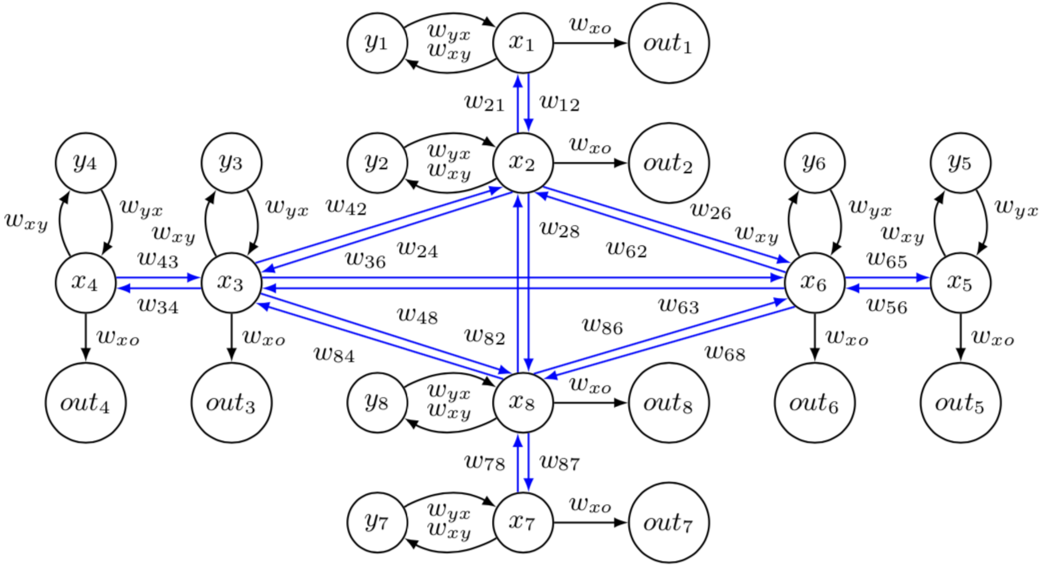




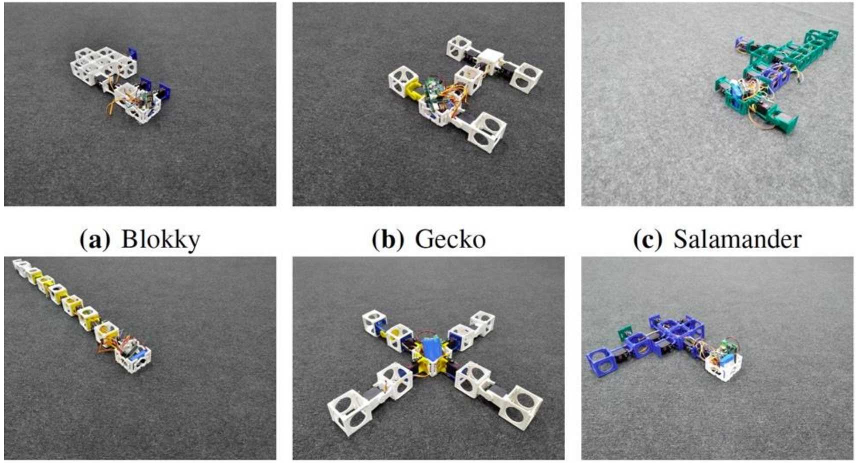
“robot zoo” in the bio-inspired robotics lab at the VU




1
2
4
3
6
5
8
7

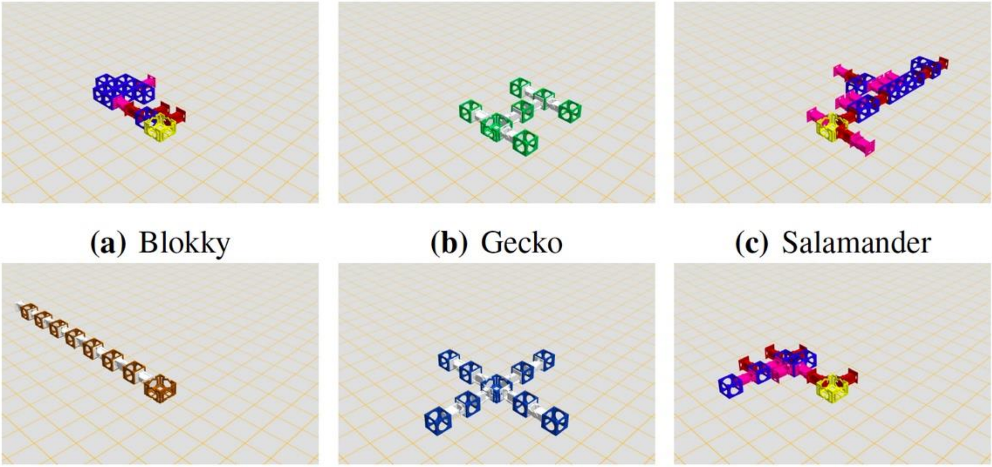

y_1
w_{yx}
w_{xy}
x_1
w_{xo}
out_1
w_{21}
w_{12}
y_2
x_2
out_2
y_4
y_3
y_6
y_5
w_{42}
w_{26}
w_{24}
w_{28}
w_{62}
w_{43}
w_{36}
w_{65}
x_4
x_3
x_6
x_5
w_{34}
w_{56}
w_{48}
w_{63}
w_{82}
w_{86}
w_{84}
w_{68}
out_4
out_3
y_8
x_8
out_8
out_6
out_5
w_{78}
w_{87}
y_7
x_7
out_7

Machine Learning → Learning Machines



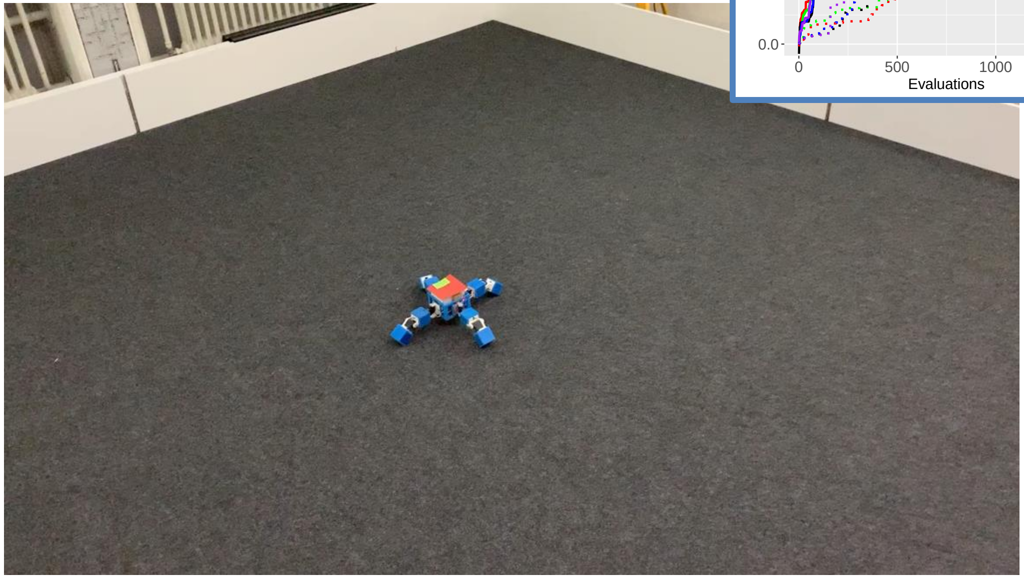
(a) Blokky (b) Gecko (c) Salamander

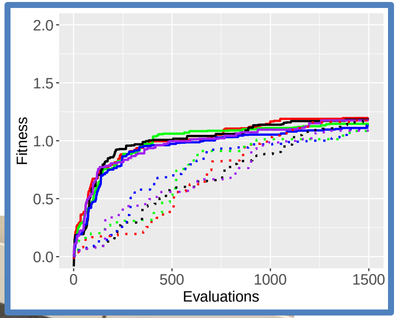
(d) Snake (e) Spider (f) Stingray



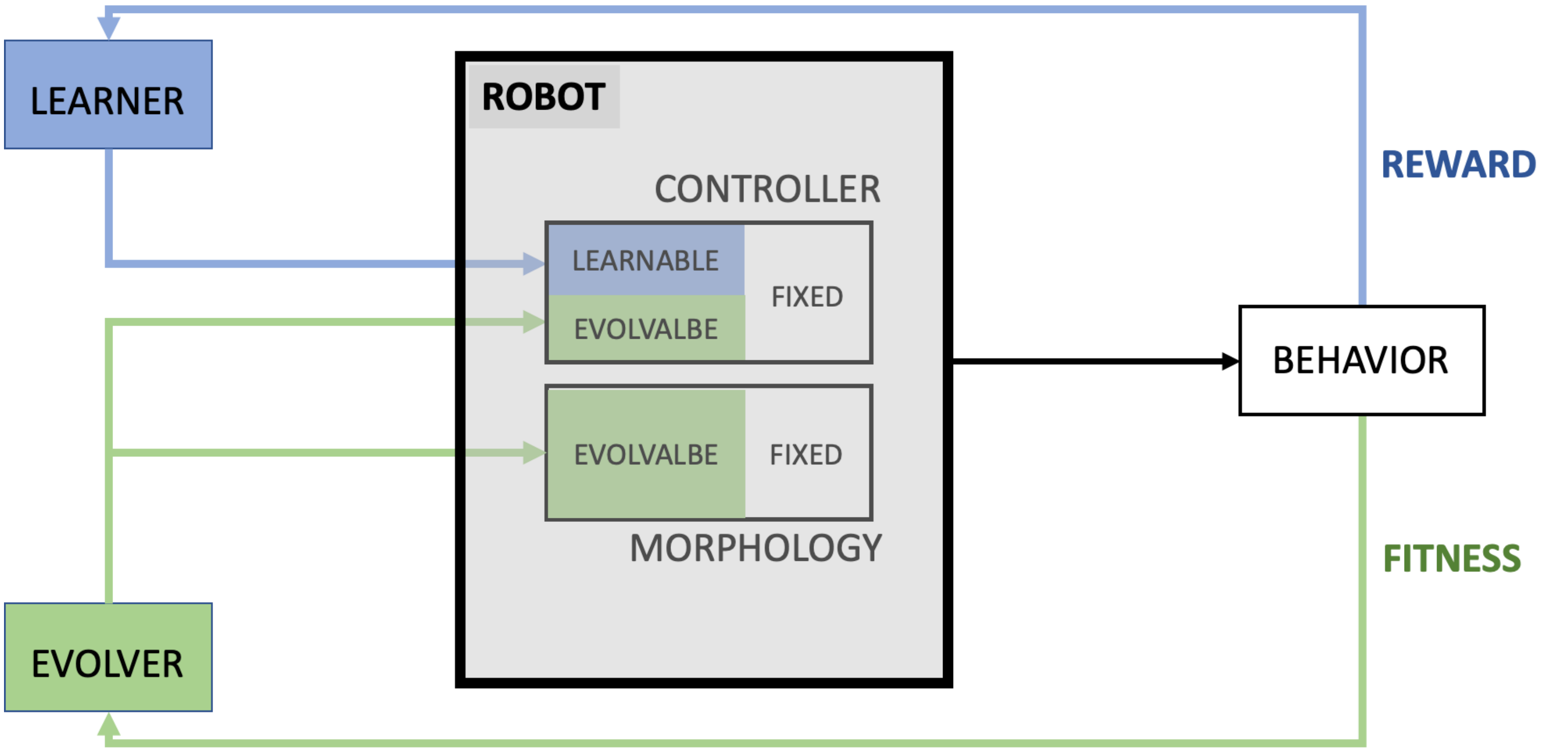
(a) Blokky (b) Gecko (c) Salamander

(d) Snake (e) Spider (f) Stingray




Fitness
Evaluations
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500

```
INITIALIZE robot population                                   [robots with body & brain]
EVALUATE each robot                                           [fitness function value]
while not STOP-EVOLUTION do
    SELECT PARENTS
    RECOMBINE + MUTATE body-coding genotypes of parents       [new body-coding genotype]
    RECOMBINE + MUTATE brain-coding genotypes of parents      [new brain-coding genotype]
    CONSTRUCT offspring phenotype                             [new robot with body & brain]
        INITIALIZE learning process
        while not STOP-LEARNING do
            TEST offspring phenotype                          [reward function value]
            GENERATE new brain for offspring
        end
    EVALUATE offspring robot with learned brain               [fitness function value]
    SELECT SUVIVORS                                           [updated population]
end
```


LEARNER
ROBOT
CONTROLLER
LEARNABLE
EVOLVALBE
FIXED
EVOLVALBE
FIXED
MORPHOLOGY
BEHAVIOR
REWARD
FITNESS
EVOLVER

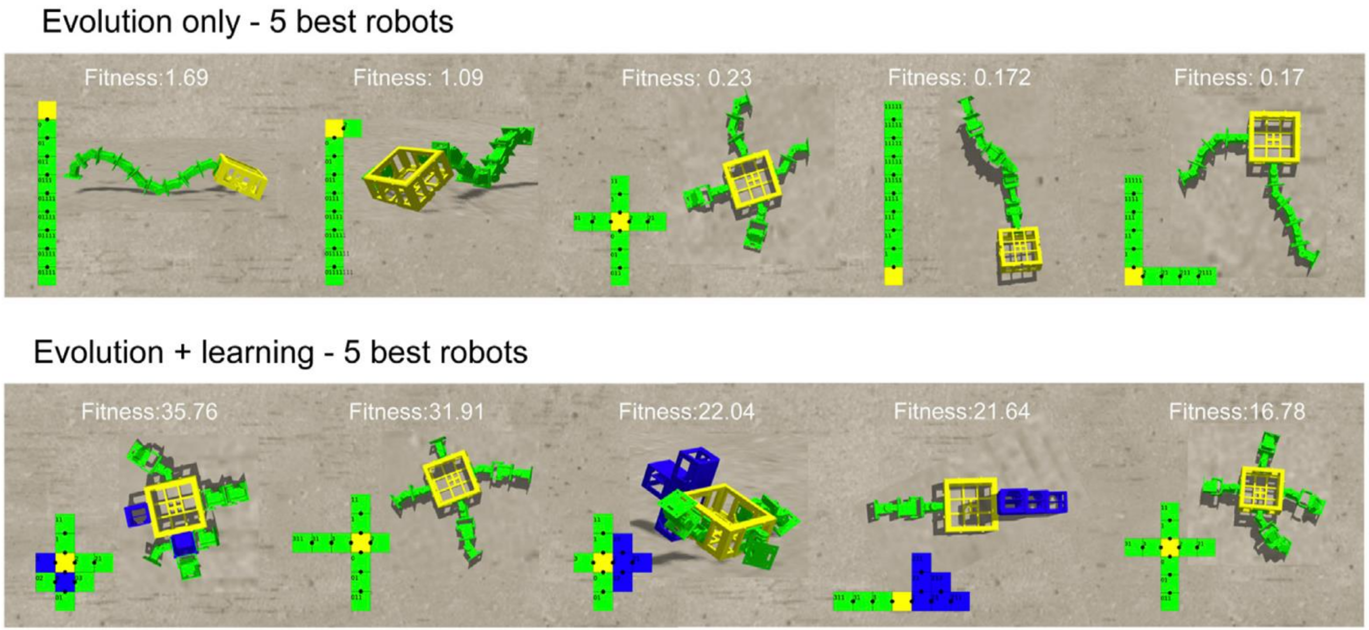
Scientific discoveries

Learning → more complex bodies

Learning delta is growing over generations
(emerging *morphological intelligence*)

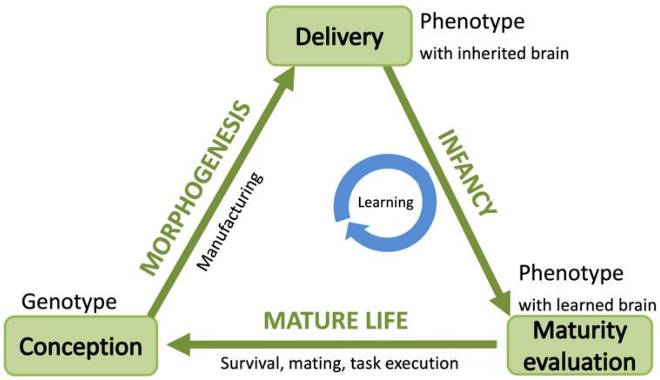
Lamarck beats Darwin

Luo, Stuurman, Tomczak, Ellers, Eiben, The Effects of Learning in Morphologically Evolving Robot Systems, *Frontiers in Robotics and AI*, 2022

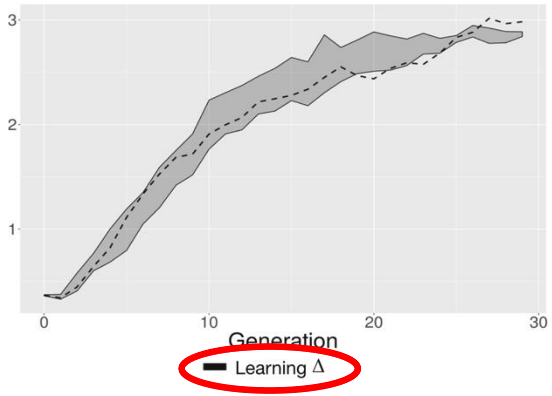


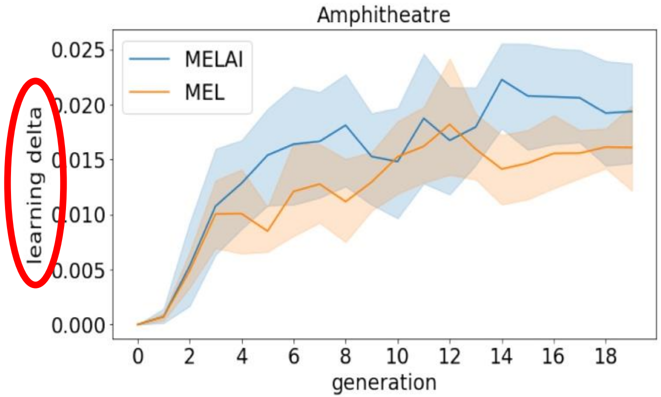
Learning delta

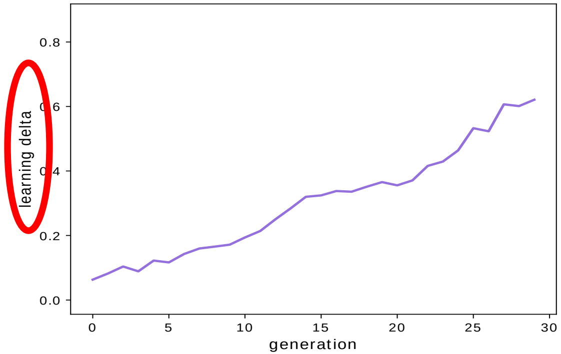
Learning delta = performance with learned brain - performance with inherited brain



Miras, De Carlo, Akhatou, Eiben, Evolving-Controllers Versus Learning-Controllers for Morphologically Evolvable Robots, ***EVOSTAR*** 2020, LNCS 12104, Springer, 2020







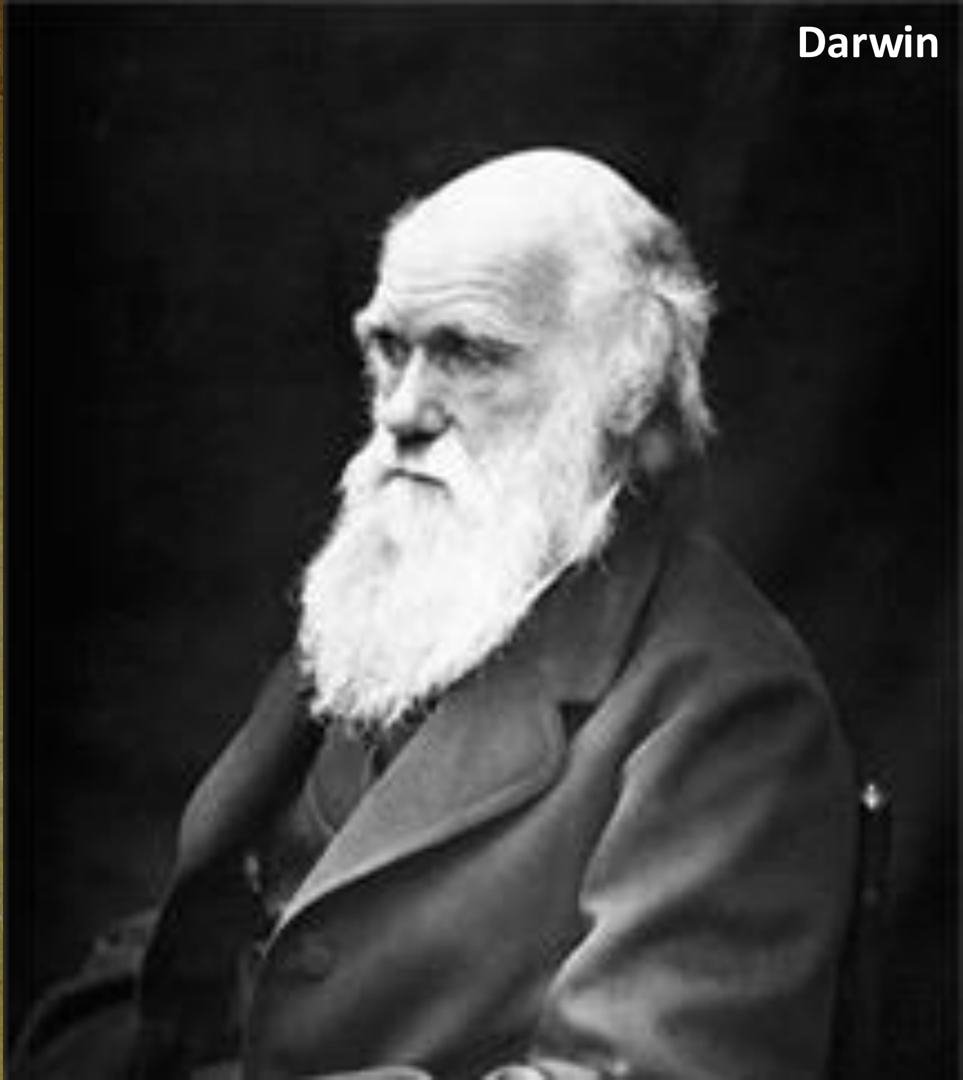
Confirmed by independently by Le Goff et al., *IEEE TCDS*, 2021

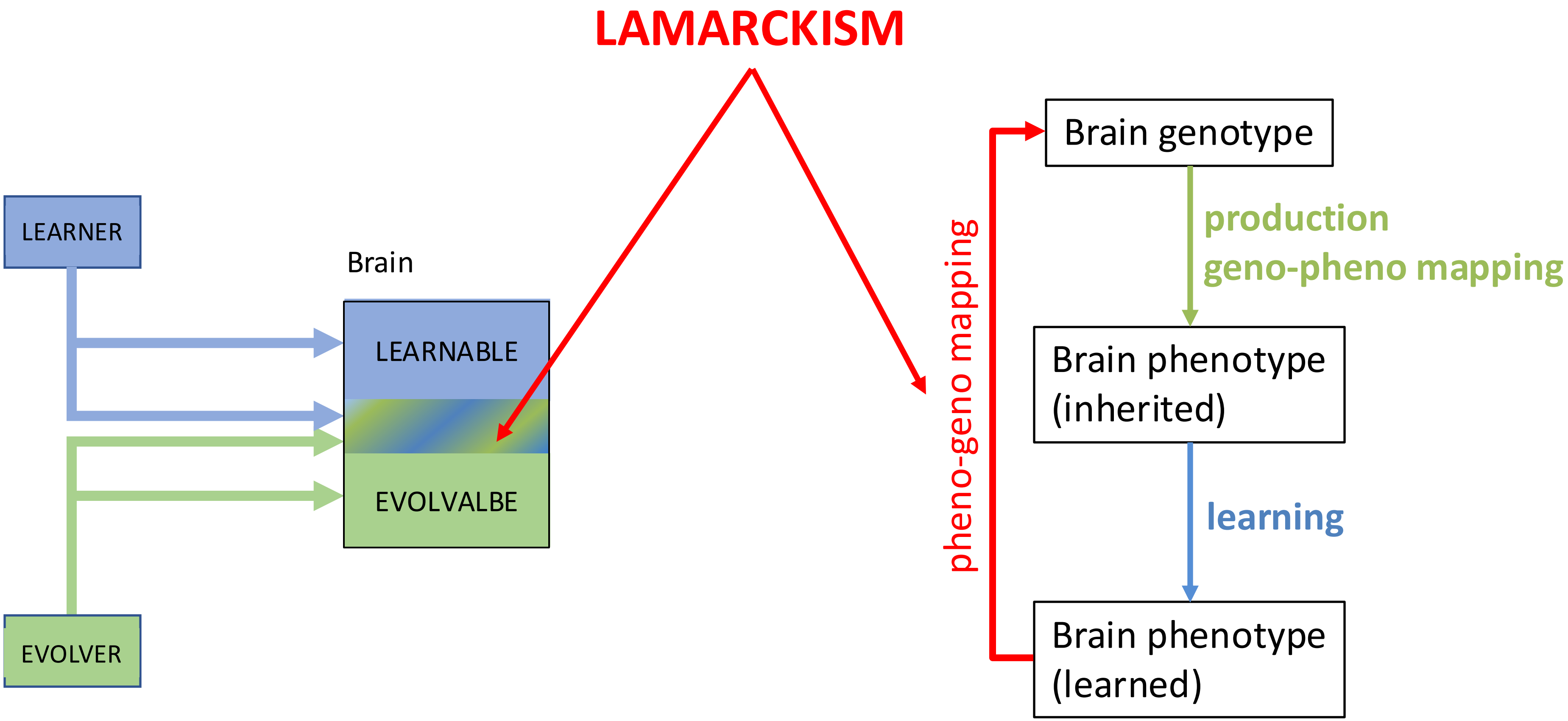
Reconfirmed by Luo, Stuurman, Tomczak, Ellers, Eiben, The Effects of Learning in Morphologically Evolving Robot Systems, *Frontiers in Robotics and AI*, 2022

Implicit selection for bodies that learn well

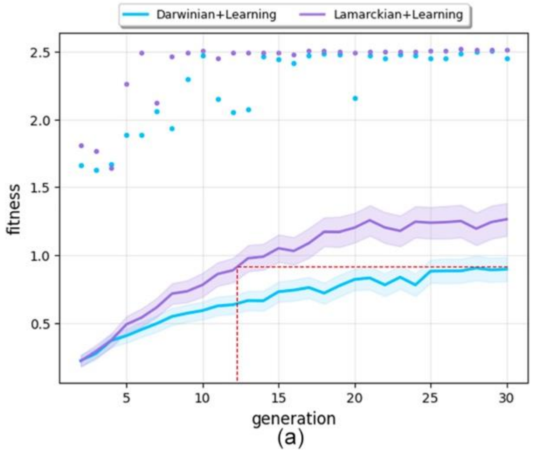
Emergence of *morphological intelligence*


Lamarck


Darwin

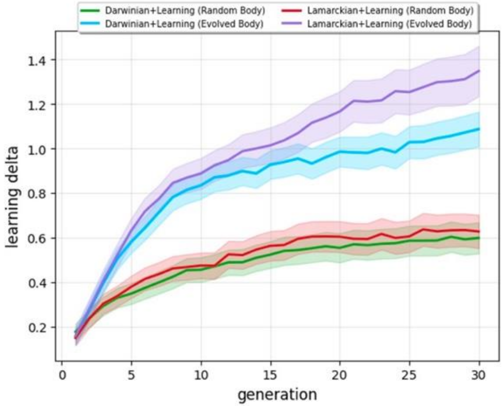

LAMARCKISM
LEARNER
Brain
LEARNABLE
EVOLVALBE
EVOLVER
pheno-geno mapping
Brain genotype
production
geno-pheno mapping
Brain phenotype (inherited)
learning
Brain phenotype (learned)

Luo, Miras, Tomczak, and Eiben, Enhancing robot evolution through Lamarckian principles, Scientific Reports, 2023



Lamarck

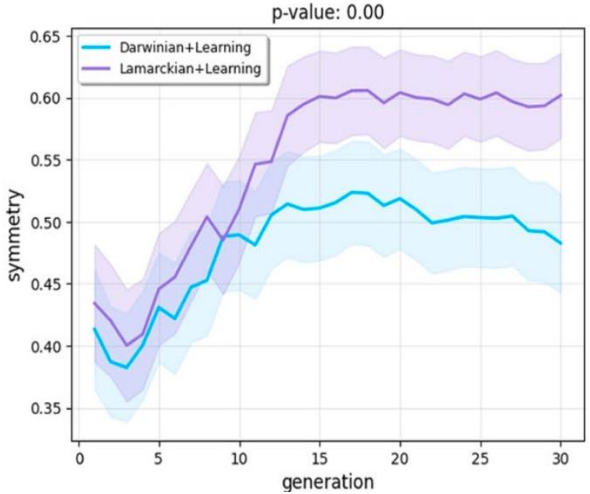
Darwin



Independently confirmed by W. Li et al., Evaluation of Frameworks That Combine Evolution and Learning to Design Robots in Complex Morphological Spaces, IEEE Transactions on Evolutionary Computation, 2023

Luo, Miras, Tomczak, and Eiben, Enhancing robot evolution through Lamarckian principles, Scientific Reports, 2023

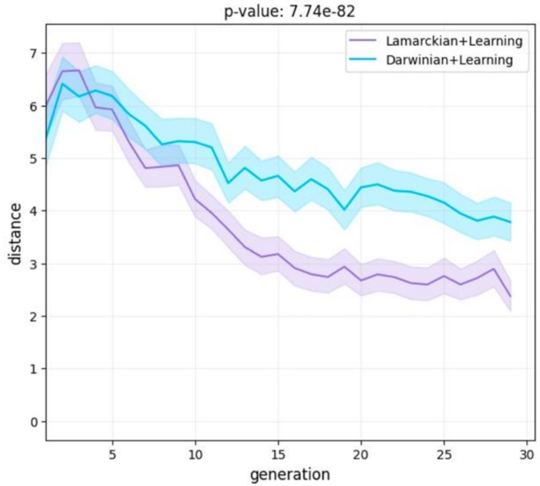
Body symmetry



Lamarck

Darwin

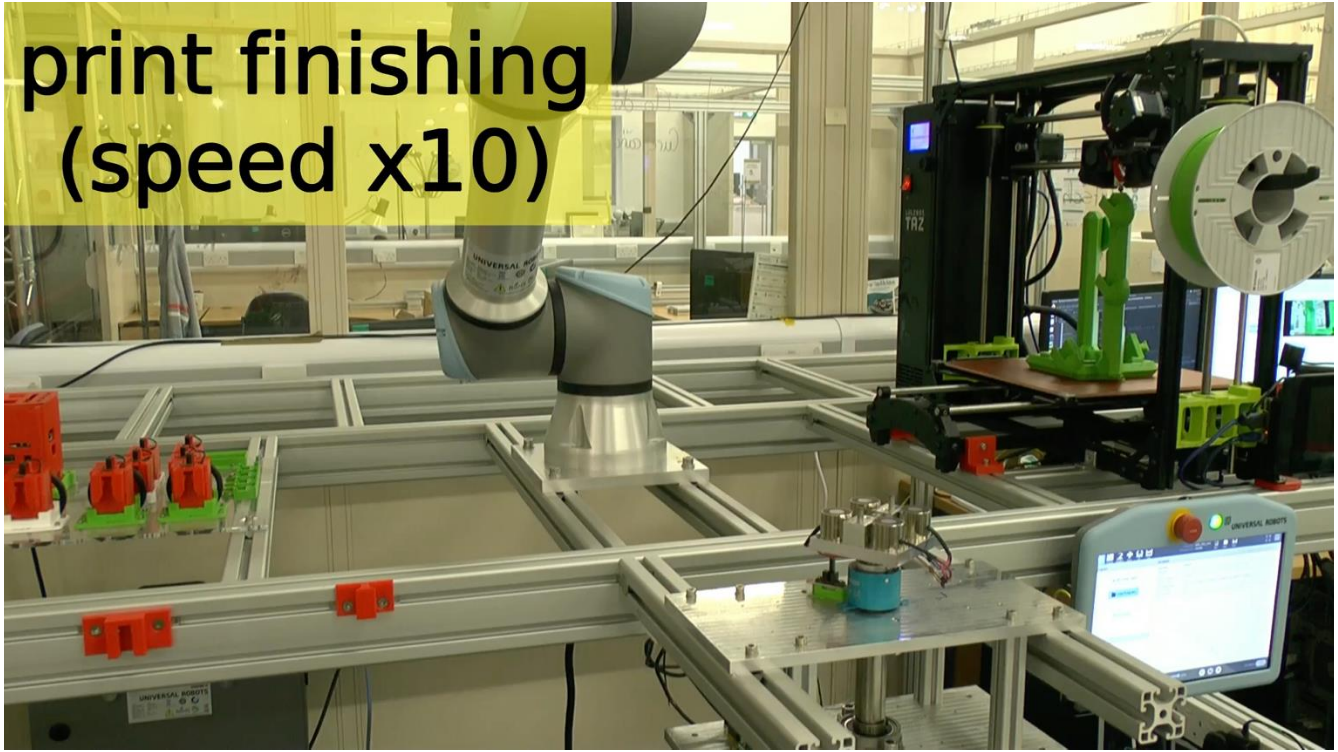
Parent-offspring distance

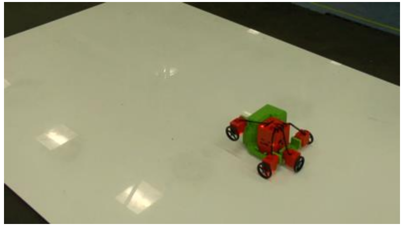


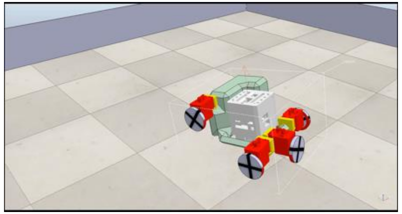
Darwin

Lamarck

Autonomous Robot Evolution project (EPSRC, 2018-2022)

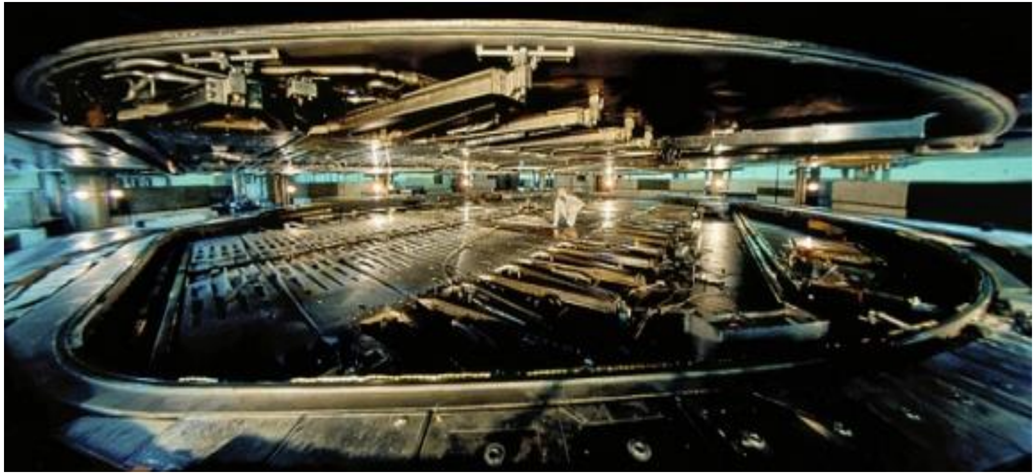






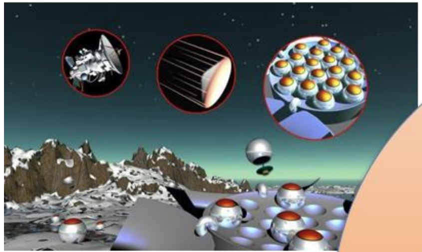
RoboFab ("Birth Clinic"), Bristol Robotics Lab

Exhibition **brAIn power**, Leiden, Museum Boerhave, Sept 2022 – March 2023



EvoSphere = novel research instrument

Stars → telescope
Particles → cyclotron
Evolution → EvoSphere



ENGINEERING

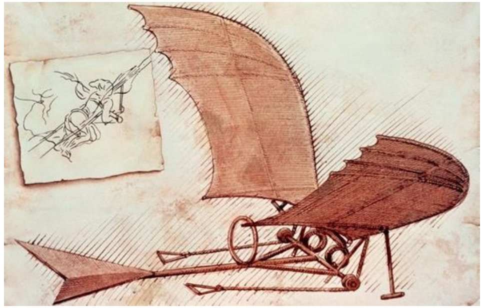
better robots

SCIENCE

fundamental research

OUTREACH

robot zoo




KLM

?

IT'S LIFE JIM

BUT NOT AS WE KNOW IT

Thanks to J. Auerbach, N. Bredeche, E. Buchanan, M. D'Angelo, M. De Carlo, F. van Diggelen, S. Doncieux , P. Eustratiadis, L. Le Goff, K. Glette, E. Haasdijk, M. Hale, E. Hart, J. Heinerman, B. Hosseinkhani Kargar, E. Hupkes, M. Hoogendoorn, D. Howard, M. Jelisavcic, S. Kernbach, R. Kiesel, G. Lan, W. Li, J. Luo, K. Miras, J.-B. Mouret , G. Nietschke, J. Orlowski, B. Paechter, C. Rossi, J. Smith, J. Stradner, A. Stuurman, J. Timmis, A. Tyrrell, B. Weel, A. Yaman, A. Winfield